Swiss UMEF
UNIVERSITY OF APPLIED SCIENCES INSTITUTE

Fintech

Payments, digital credit, blockchain, regulation — and a project: build your fintech

Swiss UMEF · Master in Finance and Banking · MFB 545 · 2026–2027

Prof. Jan Erik Meidell, PhD

Final written exam: Friday 5 February 2027, 09:45

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The complete content of the course pages, as online at janerikmeidell.com/UMEF/mfb545/ — generated on 31.10.2026.

SWISS UMEF · MASTER IN FINANCE AND BANKING · YEAR 1 · MFB 545

Fintech

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Day 1 · Fintech, or the unbundled bank	Not started	Day 2 · Payments and digital money	Not started
Day 3 · Digital credit, neobanks and financial inclusion	Not started	Day 4 · Blockchain, crypto-assets, DeFi and tokenisation	Not started
Day 5 · Workshop day · your app's full plan, on one web page	Not started	Day 6 · Strategy, funding and final presentations	Upcoming

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Welcome — here is how it works

A thirty-second read. Reopen it any time with the button at the top right.

🗓️ **Six full days**, on Friday or Saturday, 09:00–12:15 and 13:30–16:45, then the written final exam on **Friday 5 February 2027 at 09:45**.

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Got it — let's start

THE RHYTHM OF A DAY

🎤

Talk · 15 to 25 min

At the start of each block: the mechanism, the numbers, the framework. Written out in full on the page — readable on your own if the talk does not take pla

🛠️

Workshop · 45 to 55 min

You work: simulators, verified calculations, research on real players, APIs in test mode, role plays. Everything is saved in your notebook.

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DAY 1

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DAY 3

Sat 24 Oct · 09:00–16:45 · open

Digital credit, neobanks and financial inclusion

You do ·

03

DAY 4

Fri 30 Oct · 09:00–16:45 · open

Blockchain, crypto-assets, DeFi and tokenisation

You do ·

04

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- A score on 1,000 real loans (Colab) · threshold versus profitability
- Neobank economics: CAC, LTV, churn · Revolut versus N26
- A campaign, 2,000 simulated years of P2P loans · the BNPL debate
- Digital credit in Africa: effects and abuses

🚀 **Project: The pension-savings plan: payments, fees, capital at 65**

- Build your blockchain: every student is a node
- The DefiLlama tour: decentralised finance in numbers, live
- DeFi labs: Aave-style lending, Uniswap-style exchange, stablecoins
- Tokenisation, SDX, Project Helvetia, WAEMU · "Should this be a blockchain?"
- MEV, three hacks, MiCA · DeFi Millionaire

🚀 **Project: The technology choice, argued ·**

DAY 5

Sat 31 C

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Fri 5 Feb

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50% of the grade

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📘 The whole course as one PDF (cover, programme, every day, every section) · 📄 **Each day as a PDF** (every section, to read offline or print): Day 1 · Day 2 · Day 3 · Day 4 · Day 5

REVIEW · THE CLASSROOM GAMES

💰 Who Wants to Be a Billionaire?

Days 1 to 3 in thirty questions. Two teams take turns, fifteen rungs up to a billion francs, three lifelines each: 50:50, the other team, the notebook.

Open the game (room screen) ↗

🎲 Je

Day 1 … first hand … Open t…

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Saving… the ca… return… … and … 3 …sing

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⛓ B… class…

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Open the workshop

Read the sheet

What is a hash? Explained simply (in French)

The fingerprint of a piece of information: compute your own text's, watch one letter

⛓ Blockchain Demo · Anders Brownworth

A hash, a block, a chain, a distributed chain, tokens: change one piece of data and watch

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change everything, and “mine” a block like Bitcoin.

Read the sheet

the following blocks turn invalid. External site.

Open the demo ↗

THE PROJECT · “BUILD YOUR FINTECH”

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- Da…
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- Day 4 — The technology choice, argued · proving the money is there
- **Day 5** — One web page per team: the full plan, online and handed in by 16:45
- **Day 6** — The funding plan · the presentation
- **20 Nov** — **Report** (≤ 15 pages), 23:59, handed in on this site

A GOOD STARTING POINT

A specific customer: who saves, how much they can pay in each month, what they do today for their old age (a tontine, cash, their children), and **what would make them trust you**. The

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technology is chosen on Day 4.

THE TEAMS · CHOOSE YOURS

Join your team and give it a name: the list below is the one the whole class sees. Each team has its own message thread, just between its members.

TEAM 1

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course must be retaken. Every absence must be justified; **two late arrivals count as one absence.**

50% · Final exam — Friday 5 February 2027, 09:45

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Individual, in the exam room, two hours, covering the whole course: comprehension questions, numerical exercises on payments, credit and platform economics, a case analysis. Day 6 gives the instructions and sample questions.

PREPARING FOR IT

The verified calculations from the workshops (interchange, cost of a transfer, ARPU and break-even, scoring) are in the style of the exam exercises.

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- Narayanan et al. (2016). Bitcoin and Cryptocurrency Technologies ↗, chapters 1–3.
- Bonneau and Verdier (2017). *Fintech et droit* (in French). RB Édition.
- Legal texts: FINMA (fintech licence, art. 1b Banking Act; FINMA Circular 2016/7), FinSA, FinIA, AMLA · EU: PSD2 (2015/2366), MiCA (2023/1114), DORA (2022/2554) · BCEAO: instructions on electronic money and payment services · GSMA, *State of the Industry Report on Mobile Money*.

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DAY 1 · FRIDAY 16 OCTOBER 2026 · MFB 545

Fintech, or the unbundled bank

A universal bank sells six services in a single bundle: account, payment, credit, savings, investment and compliance. Fintechs, bigtechs and decentralised protocols take them over one by one. This day takes the bundle apart: your own account openings examined step by step, the unbundled value chain, a radar of a dozen Swiss, European and African players, the numbers of Wave and TWINT redone by hand — then the launch of your project.

🕘 **09:00–12:15 · 13:30–16:45 · four blocks of 90 min** | 💻 **laptop · 📱 phone**

🎤 **the talk is short — 🛠️ most of the time, you are the one working**

BY THE END OF DAY 1, YOU WILL BE ABLE TO

- Describe the six building blocks of the banking value chain, what each one earns, and why new entrants attack first the ones that require the least capital.
- Explain, step by step, how a neobank identifies a customer without a branch (reading the ID document, liveness detection, face matching, screening, declarations).
- Document a fintech player on a standard profile: problem solved, customer, revenue, funding, regulator.
- Calculate an average revenue per user (ARPU), a contribution margin and a break-even point, and compare two payment models.
- Formulate, in a team of three, the real customer problem your fintech will solve.

BEFORE YOU START — TICK

- ☐ **Signed in** on this site (the button at the bottom right): what you do today follows you and reaches the instructor.
- ☐ **A spreadsheet open** (Excel or Google Sheets) for Block 3, and a browser for the research in Block 2.
- ☐ **A Google account** created (Google Colab is used on Day 3). Nothing to install.

BLOCK 1 · 09:00–10:30 the unbundled bank · your account openings

1.1 **The unbundled bank**

Six building blocks · why the bundle is coming apart · eKYC in seven steps

🎤 TALK · 25 MIN

1.2 **Opening an account · unbundling a bank**

Your two account openings, step by step · thirteen players, six blocks

🛠 WORKSHOP · 50 MIN

BLOCK 2 · 10:45–12:15 the fintech radar · the quiz

1.3 **Six segments, one standard profile**

The radar · where to find each piece of information

🎤 TALK · 15 MIN

1.4 **“Fintech radar” workshop**

One segment per pair · three players: Swiss, European, African

🛠 WORKSHOP · 45 MIN

1.5 **End-of-morning quiz**

Eight questions · launched in class · counts toward participation

✅ QUIZ · 10 MIN

BLOCK 3 · 13:30–15:00 business models · Wave versus TWINT

1.6 **Business models**

The canvas · five revenue mechanisms · network effects · unit economics

🎤 TALK · 20 MIN

1.7 **Wave versus TWINT**

ARPU, margin, break-even · the simulator

🛠 WORKSHOP · 55 MIN

BLOCK 4 · 15:15–16:45 the “Build your fintech” project · Jeopardy

1.8 **The “Build your fintech” project**

The deliverable, the common thread of the six days, a good problem

🎤 TALK · 15 MIN

1.9 **Your team, your problem**

Teams of three · 2-min pitch · cross-critique · Jeopardy

🛠 WORKSHOP · 35 MIN

← Course page

reset my Day 1 progress on this device

SECTION 1.1 · BLOCK 1

🎤 TALK · 25 MIN

The unbundled bank

Talk by the instructor. If it does not take place, read this section on your own, then do the classification at the bottom of the page.

WHAT A BANK DOES

Six building blocks, sold in a single bundle — and what each one earns

BLOCK 1

Account

Holding the money, keeping the ledger. Earns little in fees; gives access to cheap deposits.

BLOCK 2

Payment

Moving money around: card, transfer, foreign exchange. **Very profitable** (fees, FX margin), almost without capital.

BLOCK 3

Credit

Lending out the deposits: *maturity transformation*. Earns the interest margin; requires capital and a licence.

BLOCK 4

Savings

Savings accounts, retirement provision (the 3rd pillar in Switzerland). Earns the difference between what the bank pays and what it invests.

BLOCK 5

Investment

Advice, wealth management, brokerage. Earns management fees (as a % of assets) and brokerage fees.

BLOCK 6

Compliance

Identifying the customer (KYC), monitoring flows (AMLA), reporting. A cost — which regtechs now sell as a service.

The universal bank **cross-subsidises one block with another**: the “free” account is paid for by payment fees, the FX margin and the interest margin. A new entrant does not need to do everything: it takes *the block that is most profitable for the capital it requires* — historically payments and foreign exchange — and does it better, cheaper, on a phone.

WHY THE BUNDLE IS COMING APART NOW

Three mechanisms

- **The cost of distribution collapses.** A branch costs rent, tellers and opening hours; an app, a server. Remote account opening (below) removed the last mandatory visit to a

branch.

- **The blocks can be bought.** A start-up rents a banking licence and an account from a partner bank (**banking-as-a-service**), identification from a regtech, card processing from a processor, hosting from a cloud — each block through an *API* (Day 2). It only writes the customer journey.
- **Regulation becomes proportionate.** In Switzerland, the *fintech licence* (art. 1b of the Banking Act, 2019) allows a firm to accept deposits from the public up to 100 million francs, provided it neither lends them out nor pays interest — with lighter requirements and no deposit insurance. In the EU, the payment institution (PSD2) and electronic money statuses; in the WAEMU, the BCEAO's **electronic money issuer** licence.

WHO TAKES OVER THE BLOCKS

Three kinds of new entrants — and the bank that remains

- **Specialised fintechs**: one block, one journey, one app (Wise for foreign exchange, Wave for mobile transfers, Klarna for buy-now-pay-later).
- **Bigtechs**: Apple, Google, Alibaba, Tencent. Their advantage is the loop the Bank for International Settlements calls **DNA** — *Data* (the platform's data), *Network* (hundreds of millions of users already there), *Activities* (the financial service is grafted onto commerce, messaging, the phone). Each turn of the loop enriches the data, which improves the service, which attracts users (Frost et al., 2019).
- **Decentralised protocols**: lending, exchange, stablecoins run by programs on a blockchain, with no company in the middle (Day 4).
- **And the bank**: it keeps what requires a balance sheet — credit, large deposits — and often becomes the invisible supplier of fintechs. The opposite movement also exists: *super-apps* (Alipay, M-Pesa) re-bundle everything into a single app (Day 6).

THE MECHANISM · OPENING AN ACCOUNT WITHOUT A BRANCH

Remote identification (eKYC), step by step

In 1.2, you will compare your own account openings with these seven steps. In Switzerland, the framework is **FINMA Circular 2016/7** "Video and online identification". Each step answers a question: *is the document genuine, is the person alive, is it the same person, are we allowed to accept them?*

THE SEVEN STEPS OF AN ONBOARDING

1 · Capturing the document. The app photographs the passport or ID card. It reads the *machine-readable zone* (the two lines of chevrons "<<<" at the bottom of the photo page) by optical character recognition. Each field there carries a **check digit**: the characters are multiplied by 7, 3, 1, 7, 3, 1… (letters A = 10 … Z = 35), the products are added up, and the remainder of the division by 10 is kept. For the number of the ICAO specimen passport, L898902C3, the check digit is 6. A misread or a clumsy forgery shows up immediately.

```
L=21 8 9 8 9 0 2 C=12 3 · weights 7 3 1 7 3 1 7 3 1 →
147+24+9+56+27+0+14+36+3 = 316 → 316 mod 10 = 6
```

2 · Authenticity of the document. The security features are checked in the image: holograms that change with the tilt angle (the user is asked to move the document), microprinting, consistency of the font. Biometric passports also have a **chip** readable by NFC on some phones: it contains the data and the photo, signed by the issuing State — a signature the app can verify.

3 · Selfie and liveness detection. A photo is not enough: someone could present a photo of someone else. *Liveness* detection is *active* (turning the head, following a dot, reading digits aloud) or *passive* (a model analyses skin texture, reflections and depth in a single image).

4 · Face matching. A neural network turns each face — the one on the document and the one in the selfie — into a vector of a few hundred numbers. The closeness of the two vectors is measured; above a **threshold**, it is the same person. High threshold: fewer fraudsters accepted, more genuine customers rejected. Low threshold: the opposite. The threshold is a business choice, not a technical truth.

5 · Screening. The name and date of birth are compared with sanctions lists (SECO, UN, EU, the US OFAC) and with lists of **politically exposed persons**. Names are spelled in several ways: the matching is fuzzy, and a human decides the doubtful cases (Day 5).

6 · Declarations. The customer declares the beneficial owner of the funds (in Switzerland, "Form A" of the banks' Agreement on due diligence), their tax residence (automatic exchange of information) and whether they are a *US person* (FATCA). A questionnaire on the source of funds is added above certain amounts.

7 · The first transfer. For *online* identification (without a video call), FINMA Circular 2016/7 requires a transfer from a bank account in the customer's name: a second bank

has already identified the person, and the name must match. In *video* identification, a trained employee conducts the live call instead.

✓ All 7 steps are shown

> **The ten-minute account is no less checked than the branch one: *the checking has been automated, and each step has an error rate that can be tuned*.**

CHECK · 3 MIN

Specialised fintech, bigtech, decentralised protocol or bank?

▸ CLASSIFY EACH PLAYER

Wise — international transfers at the mid-market rate plus displayed fees

Fintech | Bigtech | Decentralised | Bank

Apple Pay — paying with your iPhone

Fintech | Bigtech | Decentralised | Bank

Uniswap — exchanging tokens on a blockchain

Fintech | Bigtech | Decentralised | Bank

PostFinance — accounts, payments, savings, investment

Fintech | Bigtech | Decentralised | Bank

Alipay — payment built into Alibaba's platform

Fintech | Bigtech | Decentralised | Bank

Wave — mobile wallet in Senegal and Côte d'Ivoire

Fintech | Bigtech | Decentralised | Bank

Aave — lending and borrowing crypto-assets

Fintech | Bigtech | Decentralised | Bank

0 of 7 sorted

SECTION 1.2 · BLOCK 1

🛠 WORKSHOP · 50 MIN

Your account openings · unbundling a bank

A · 20 MIN · ALONE, THEN IN PAIRS

Your own account openings, examined against the seven steps

Everyone has already opened at least one account: in Switzerland on arrival (a bank, PostFinance, an app), and in their home country (a bank account, Wave, Orange Money, MTN MoMo…). **Choose two of these journeys** and reconstruct them from memory, step by step, using the seven eKYC steps seen in 1.1. Nothing to open, no personal data to enter: only what you remember. **10 minutes alone**, then **10 minutes in pairs**: compare your tables and find the step that took the longest.

STEP	Journey 1 — e.g. bank in	Journey 2 — e.g. Wave, D
Appointment or mandatory branch visit	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
1 · ID document photographed or read	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
2 · Authenticity of the document checked	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
3 · Selfie or video call	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
4 · Face compared with the document	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
5 · Screening (sanctions, exposed persons) — often invisible	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
6 · Declarations: beneficial owner, tax residence, source of funds	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
7 · First deposit or transfer required	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know

0 of 16 cells filled in

Copy the table into my notebook ↓

1. Your two journeys: the table, then for each one the total duration (from first contact to the first possible payment), the documents requested and the fees.

The “Copy the table” button inserts your answers; complete duration, documents and fees.

▸ THE TYPICAL JOURNEY AT A TRADITIONAL BANK, FOR COMPARISON

STEP	WHERE	TYPICAL TIME
Book an appointment	phone, website	a few days
Interview, original ID document copied, Form A (beneficial owner), tax declarations	branch	45–60 min on site
Internal checks, account opened	head office	1–3 working days
Card and codes sent by post	mail	5–10 working days

Typical journey based on the “open an account” pages of Swiss banks; it varies by institution and customer profile. The legal steps are the same as in the app: only the way they are carried out changes.

2. In your two journeys, where was the friction? Which of the seven steps could an app have removed or sped up — and which not?

e.g. In Switzerland, it took a proof of address and a residence permit, three weeks; at Wave, an ID document at the agent’s and ten minutes…

B · 30 MIN · “BANKING VALUE CHAIN” SIMULATOR

Thirteen real players. For each one, the bank's building block it has taken over.

Look up the ones you do not know (company website, two minutes each). Several do a bit of everything: choose *the block that generates their revenue*. A wrong answer greys out the button; the explanation appears when you are right.

ACCOUNT · PAYMENT · CREDIT · SAVINGS · INVESTMENT · COMPLIANCE

TWINT (Switzerland)

Account | Payment | Credit | Savings | Investment | Compliance

Wise (United Kingdom)

Account | Payment | Credit | Savings | Investment | Compliance

Flutterwave (Nigeria)

Account | Payment | Credit | Savings | Investment | Compliance

neon (Switzerland)

Account | Payment | Credit | Savings | Investment | Compliance

Klarna (Sweden)

Account | Payment | Credit | Savings | Investment | Compliance

swisspeers (Switzerland)

Account | Payment | Credit | Savings | Investment | Compliance

Baobab (Africa, formerly MicroCred)

Account | Payment | Credit | Savings | Investment | Compliance

VIAC (Switzerland)

Account | Payment | Credit | Savings | Investment | Compliance

frankly (Switzerland, Zurich Cantonal Bank)

Account | Payment | Credit | Savings | Investment | Compliance

True Wealth (Switzerland)

Account | Payment | Credit | Savings | Investment | Compliance

Trade Republic (Germany)

Account | Payment | Credit | Savings | Investment | Compliance

PXL Vision (Switzerland)

Account | Payment | Credit | Savings | Investment | Compliance

Smile ID (Africa)

Account | Payment | Credit | Savings | Investment | Compliance

0 of 13 sorted

DEBRIEF · 15 MIN

Three questions for the room: which block has the most new entrants, and why? Which one has almost none? What does the universal bank still do that nobody has taken from it?

SECTION 1.3 · BLOCK 2

🎤 TALK · 15 MIN

Six segments, and how to document a player

Talk by the instructor, or self-study.

THE RADAR'S SEGMENTS

The market cut into six — with, for each, the mechanism that makes money

SEGMENT 1

Payments

Fee per transaction, FX margin. *TWINT, Wise, Stripe, Wave, Flutterwave.*

SEGMENT 2

Credit

Interest margin or merchant fee (buy-now-pay-later). *Klarna, swisspeers, Baobab.*

SEGMENT 3

Savings and wealth management

Management fees as a % of assets, brokerage fees. *VIAC, True Wealth, Yuh, Trade Republic.*

SEGMENT 4

Insurtech

Premiums, distribution commissions; often a traditional insurer's policy sold differently. *Lemonade, Wefox, Turaco.*

SEGMENT 5

Regtech

Subscription or price per check, sold to institutions. *PXL Vision, IDnow, Smile ID.*

SEGMENT 6

Infrastructure and BaaS

Price per account or per API call, sold to other fintechs. *Solaris, Swan, Onafriq, Paystack.*

The names are starting points, not a closed list: a player you know from your own country is worth more than a name from this page.

THE STANDARD PROFILE

Five questions, and where to find each answer

- **Problem solved** — in one sentence, from the customer's point of view, *without* the name of the technology. Source: the home page, then customer reviews (app stores).
- **Customer** — who pays, and who uses (not always the same: at TWINT the user pays nothing, the merchant pays). Source: the price list.

- **Revenue** — the exact mechanism: “1% on transfers”, “0.5% of assets per year”, “CHF 9 per month”. Source: the pricing terms, the annual report if published.
- **Funding** — funds raised, investors, profitable or not. Source: press releases, specialised press (Finextra, TechCabal, Fintech News Switzerland), Crunchbase.
- **Regulator** — who authorises it, under which status. Source: the regulator’s register (FINMA: list of authorised institutions; BCEAO: list of electronic money issuers; EU: national registers and the EBA register), and the website’s legal notice.

> **A profile is judged on its revenue mechanism. *“Makes money from fees” is not an answer; “charges 1% on every transfer, deposits and withdrawals free” is one.***

CHECK · 3 MIN

Where to look?

› WEBSITE AND PRICING · REGULATOR’S REGISTER · PRESS AND RELEASES · ANNUAL REPORT

How much it costs to send 100,000 FCFA with a mobile money provider

Website and pricing | Regulator’s register | Press, releases | Annual report

Whether a Swiss fintech has a banking licence, a fintech licence, or nothing

Website and pricing | Regulator’s register | Press, releases | Annual report

How much a fintech raised in its last funding round, and from whom

Website and pricing | Regulator’s register | Press, releases | Annual report

The revenue and profit of a listed fintech

Website and pricing | Regulator’s register | Press, releases | Annual report

Whether an institution is licensed to issue electronic money in Côte d’Ivoire

Website and pricing | Regulator’s register | Press, releases | Annual report

0 of 5 sorted

SECTION 1.4 · BLOCK 2

🛠 WORKSHOP · 45 MIN

“Fintech radar” workshop

30 MIN · IN PAIRS

One segment per pair. Three real players: one Swiss, one European, one African.

Nine people: three pairs and one group of three. Each group takes a different segment from the six in 1.3 (the two remaining segments are for groups that finish early). Fill in one profile per player: the button inserts the template into an empty field. Everyone fills in their own three profiles — it is your notebook, and it is saved.

Insert the template into the empty fields

Swiss player — segment, name, and the five lines of the profile

European player

African player

15 MIN · REPORT-BACK AND COLLECTIVE RADAR

Three minutes per group. The radar is built on the screen.

Each group presents its segment in three minutes: the three players, and *one* thing they have in common (the same revenue mechanism? the same customer?). The whole room's profiles are shown on the screen, anonymously, during the report-back.

Report-back 2:55 Pause ↺

SECTION 1.5 · BLOCK 2

✅ QUIZ · 10 MIN

End-of-morning quiz

INTERACTIVE QUIZ · COUNTS TOWARD PARTICIPATION

Eight questions, one attempt each, all explained

Launched in class, so that everyone starts at the same time; this page unlocks by itself. Interactive quiz results count toward the 10% participation mark; a missed quiz scores zero unless arranged in advance. On your own, without an AI assistant.

0 / 8 answered

1. “Unbundling” the universal bank means…

- merging several banks into one
- closing branches in favour of an app
- having specialised players offer separately services that the bank used to sell in a single bundle
- abolishing banking regulation

2. The Swiss fintech licence (art. 1b of the Banking Act) allows a firm…

- to accept deposits from the public up to 100 million francs, without lending them out or paying interest
- to lend out the deposits received without limit
- to issue banknotes
- to operate without any supervision

3. In Switzerland, opening an account by video or online is governed by…

- the GDPR

- the PSD2 directive
- the MiCA regulation
- FINMA Circular 2016/7

4. neon offers accounts and cards in Switzerland. How, without a banking licence?

- It does not need one to hold accounts
- The accounts are held by a partner bank
- It uses the fintech licence to lend
- The funds are placed on a blockchain

5. A two-sided network effect is…

- a service that costs half as much at large scale
- a network of two banks
- an app available on two operating systems
- two groups that attract each other: the more customers pay with an app, the more merchants accept it, and vice versa

6. ARPU of 500 FCFA per month, variable costs of 250 FCFA per user, fixed costs of 1.5 billion FCFA per month. The break-even point is at…

- 6 million active users
- 3 million
- 12 million
- 1.5 million

7. Wave's pricing model in Senegal is…

- a fixed monthly subscription
- tiered, decreasing fees on each operation
- free deposits and withdrawals, 1% on transfers
- free transfers funded by advertising

8. Which of these players belongs to regtech?

- A low-fee online broker
- A provider of identity verification and sanctions-list screening sold to banks
- A payment app between friends
- A crowdfunding platform

SECTION 1.6 · BLOCK 3

🎤 TALK · 20 MIN

Business models: who pays, how much, and when it becomes profitable

Talk by the instructor, or self-study.

THE CANVAS

Nine boxes to describe how a company makes money

The *Business Model Canvas* (Osterwalder and Pigneur, 2010), filled in here for Wave in broad strokes. On the right, value and revenue; on the left, what it takes to produce them and what it costs.

KEY PARTNERS
Banks where customers' funds are deposited · agent network · telecom operators

KEY ACTIVITIES
App, transaction processing, recruiting and managing agents, compliance

KEY RESOURCES
Electronic money issuer licence, software, liquidity at the agents

VALUE PROPOSITION
Send money from a phone for 1%, deposit and withdraw for free

CUSTOMER RELATIONSHIPS
Simple app, QR card for those without a smartphone, customer service

CHANNELS
Neighbourhood agents, word of mouth, app

CUSTOMER SEGMENTS
Unbanked or underserved individuals · merchants · companies paying salaries

COST STRUCTURE
Agent commissions · technology · salaries · marketing · compliance

REVENUE STREAMS
1% on transfers · services to merchants and businesses · income on deposited funds, within regulatory limits

HOW A FINTECH CHARGES

Five revenue mechanisms

- **Commission or fee per transaction** — a percentage or a fixed amount on each operation (merchant fee, transfer fee).
- **Interchange** — the share of each card payment received by the bank that issued the card. This is how neobanks make money on a "free" card (Day 2).
- **Margin** — on foreign exchange (converting at a worse rate than the mid-market rate) or on interest (lending at a higher rate than one's own funding cost).

- **Subscription** — a monthly price for a premium offer.
- **Management fees** — an annual percentage of assets under management.

NETWORK EFFECTS

Why the first to reach critical mass often wins everything

Direct effect — each new user makes the service more useful to the others. Among n users, the number of pairs that can send each other money is $n(n-1)/2$: 45 pairs for 10 users, 4,950 for 100, 499,500 for 1,000. The value grows much faster than the number of users.

Indirect, or two-sided, effect — two groups that attract each other: the more customers pay with TWINT, the more merchants accept it, the more customers adopt it. The starting problem is the *chicken-and-egg* problem; the usual solution is to **subsidise one side** — free P2P at TWINT, free withdrawals at Wave — and charge the other.

UNIT ECONOMICS

Four formulas for Block 3

ARPU = revenue for the month ÷ active users for the month

Contribution margin per user = ARPU – variable costs per user

Break-even (in users) = fixed costs for the month ÷ contribution margin per user

Take rate = revenue ÷ volume that flows through

Variable costs follow each user (agent commissions, processing, SMS, fraud); *fixed* costs are there even with no users (salaries, technology, compliance, head office). A fintech loses money as long as it is below its break-even point, and each user above it brings in their contribution margin.

CHECK · 3 MIN

Which revenue mechanism?

PER TRANSACTION · INTERCHANGE · MARGIN · SUBSCRIPTION · MANAGEMENT FEES

The merchant pays 1.3% of each purchase paid with the app

Per transaction | Interchange | Margin | Subscription | Management fees

A neobank gives the card away; it receives a small share of each payment made with it

Per transaction | Interchange | Margin | Subscription | Management fees

A bank converts your francs into euros at 0.91 while the mid-market rate is 0.93

Per transaction | Interchange | Margin | Subscription | Management fees

A “premium” offer at CHF 9 per month with travel insurance and free withdrawals

Per transaction | Interchange | Margin | Subscription | Management fees

A robo-advisor charges 0.5% of assets per year

Per transaction | Interchange | Margin | Subscription | Management fees

An online lender lends at 12% and funds itself at 5%

Per transaction | Interchange | Margin | Subscription | Management fees

Wise displays a fee of CHF 4.20 on a transfer, at the mid-market rate

Per transaction | Interchange | Margin | Subscription | Management fees

0 of 7 sorted

SECTION 1.7 · BLOCK 3

🛠 WORKSHOP · 55 MIN

Wave versus TWINT — redoing the numbers

THE DATA · PER ACTIVE USER PER MONTH

Two payment apps, two continents, two models

	WAVE (SENEGAL) · FCFA	TWINT (SWITZERLAND) · CHF
Person-to-person transfers (P2P)	50,000 · fee **1%** for the sender	80 · **free**
Payments at merchants	— (not included)	120 · merchant fee **1.3%**
Cash deposits and withdrawals at agents	60,000 · free for the customer; Wave pays the agent **0.3%**	— (no cash)
Other variable costs (processing, SMS, support, fraud)	70	0.46
Company fixed costs, per month	1.5 billion	5.5 million
Active users	8 million	6 million

⚠ **Stylised figures for the exercise**, built on public orders of magnitude (Wave's "1% on sending, deposits and withdrawals free" pricing; a merchant fee of around 1.3% for TWINT; free P2P) and rounded so the calculations can be done by hand. These companies' real figures are not published in detail: do not quote these as such.

25 MIN · BY HAND, THEN IN THE SPREADSHEET

Five calculations

Do them in your spreadsheet: one column per company, one row per quantity. Enter each result here; a mistake can be corrected as many times as needed.

Wave · ARPU. Revenue per active user per month (FCFA).

… FCFA Check

Wave · contribution margin per user per month (FCFA).

... FCFA Check

Wave · break-even in active users.

... users Check

TWINT · break-even in active users.

... users Check

Price war. A competitor forces Wave to go from 1% to 0.8% on transfers. New break-even point, in active users?

... users Check

10 MIN · VARY THE INPUTS

The simulator: move the sliders, watch the break-even point move

Wave | TWINT

Fee on transfers 1.0 %

Transfers per user per month 50,000

Agent commission (on 60,000 of deposits/withdrawals) 0.30 %

Fixed costs per month (billions) 1.5

Active users (millions) 8.0

500	250	6.00 M	0.50 bn
ARPU · FCFA	MARGIN / USER	BREAK-EVEN · USERS	RESULT FOR THE MONTH · FCFA

Above break-even: each additional user adds their margin to the result.

10 MIN

Wave, TWINT or both?

▸ CLASSIFY EACH FEATURE

Sending money between individuals is free for the user

Wave | TWINT | Both

Depositing and withdrawing cash is free

Wave | TWINT | Both

Relies on a network of physical agents

Wave | TWINT | Both

Debits a bank account directly

Wave | TWINT | Both

Network effect between users and merchants

Wave | TWINT | Both

Payment by QR code

Wave | TWINT | Both

Forced a telecom operator to cut its prices sharply

Wave | TWINT | Both

0 of 7 sorted

✍ **In these data, Wave's ARPU is about CHF 0.70, against CHF 1.56 for TWINT. In three sentences: how can Wave still be profitable, and what would make it fragile?**

e.g. Its fixed costs in CFA francs are low for…; it exceeds its break-even point by…; but cutting its fee to 0.8% would move the break-even point to…

SECTION 1.8 · BLOCK 4

🎤 TALK · 15 MIN

The “Build your fintech” project

Talk by the instructor, or self-study.

40% OF THE GRADE · REPLACES THE MIDTERM EXAM

Two teams of three, one project: a pension-savings app for Senegal, Congo and their neighbours.

The app lets people who have no pension, or a very small one, save for their old age by paying **small amounts every month until 65**, invested in **Swiss blue-chip equity funds**, among the most stable on the market. The app and the company must be **transparent and earn trust**: that is the condition for anyone to hand over their money for thirty years. Each team picks its country (Senegal, Congo, DRC, Côte d’Ivoire…) and its customer. Each day adds a piece to the project, at the end of the afternoon, using what has just been learned. *(Brief set on 24 October 2026.)*

Day 1 · 16 Oct.

The problem and the customer — the team, the market, the problem in one sentence

Day 2 · 23 Oct.

The payment flow and the revenue model

Day 3 · 24 Oct.

The pension-savings plan — payments, fees, capital at 65, transparency

Day 4 · 30 Oct.

The technology choice — is a blockchain needed to prove the money is there? a reasoned decision

Day 5 · 7 Nov.

The regulatory file and the risk map

Day 6 · 14 Nov.

The financing plan · **presentation to the jury** (10 min + 10 min of questions)

The deliverable: (a) a report of at most fifteen pages — problem and customer, market, the savings plan and the customer journey with a clickable prototype, revenue model and unit economics, the payment flow to the Swiss fund and back, what creates trust (custody of the funds, transparent fees, information on risks), reasoned technology choice, regulatory compliance in Switzerland and in the target country, financing plan; (b) the prototype of the customer journey; (c) the oral presentation on 14 November. **Report due Friday 20 November 2026, 23:59**, on this site — like everything else in the project: nothing is handed in on Moodle or the Swiss UMEF platform. One grade for the whole team, weighted where appropriate by an individual assessment of each member’s contribution.

A GOOD STARTING POINT

A problem, not a solution

- **A specific customer**: “self-employed seamstresses in Dakar”, not “Africans”.
- **What they do today**: the current workaround (cash, tontine, a transfer through a cousin) — that is your real competitor.
- **What it costs them**: in money, in time, in risk. A figure, even a rough one.
- **Why now**: what has changed (a phone in every pocket, instant payments, a new regulation).

> **The word “blockchain” or “AI” does not appear in a problem statement. *The technology is chosen on Day 4, once the problem is understood.***

CHECK · 3 MIN

Good problem, disguised solution, or too vague?

▸ CLASSIFY EACH STARTING POINT

“Build a neobank for Africa”

Good problem | Disguised solution | Too vague

“An app with AI and blockchain for payments”

Good problem | Disguised solution | Too vague

“African students in Geneva often pay more than 5% to send CHF 200 to their family every month”

Good problem | Disguised solution | Too vague

“Seamstresses in Dakar have no credit history to borrow 300,000 FCFA of fabric before Tabaski”

Good problem | Disguised solution | Too vague

“Create a digital token to replace the CFA franc”

Good problem | Disguised solution | Too vague

“Swiss SMEs wait on average more than 30 days for their invoices to be paid”

Good problem | Disguised solution | Too vague

“Improve financial inclusion”

Good problem | Disguised solution | Too vague

0 of 7 sorted

SECTION 1.9 · BLOCK 4

WORKSHOP · 35 MIN | PITCH, CRITIQUE AND JEOPARDY · 40 MIN

Your team, your problem

10 MIN · FORMING THE TEAM

Two teams of three, one country each

A team should preferably mix people who do not know each other yet and who do not come from the same country. Each member fills in the field below with the same information.

1. Team name · the three members · the country your pension-savings app targets (Senegal, Congo, DRC, Côte d'Ivoire… — and which city first)

e.g. Team Teranga · A., B., C. · Senegal, Dakar

25 MIN · THE PROBLEM

The problem in one sentence, and the customer in three lines

2. The problem: *For [specific customer], who today [does it how], the problem is [which one], which costs them [how much].*

e.g. For women traders aged 25 to 45 in Dakar, who save today in cash or in a tontine, the problem is that they will have no pension at 65 and their savings earn nothing, which costs them…

3. The customer: who exactly, how many are there (order of magnitude and source), and how do you know the problem exists?

e.g. About … students (source: …); two team members live this problem; three people interviewed during the break…

20 MIN · TWO-MINUTE PITCH AND CROSS-CRITIQUE

Each team presents; the other asks its three questions

Pitch 1:55 Pause ↺

The listening team asks, in this order: **1.** Does the customer really exist, and have you met them? **2.** What do they do today, and why would they change? **3.** What could kill this project (a competitor, regulation, acquisition cost)?

4. The critique received, and what the team changes in its problem.

e.g. We were asked whether students do not already use Wise; we will compare the costs on Day 2…

20 MIN · CLASS GAME

Jeopardy “Fintech — fundamentals”

The three project teams play against each other on the classroom screen: five categories, twenty-five questions on the day. A team picks a square; the first raised hand answers.

Open the Jeopardy (classroom screen) ↗

BEFORE DAY 2 · FRIDAY 23 OCTOBER

- Refine the problem with the critique received; interview two potential customers.
- Day 2 is about payments: note how your customer pays and receives money today — by what means, at what cost, how quickly.

DAY 2 · FRIDAY 23 OCTOBER 2026 · MFB 545

Payments and digital money

A card payment involves five parties in under two seconds; a mobile money withdrawal in Dakar depends on the liquidity of a neighbourhood agent; a CHF 200 transfer costs anything from under 1% to over 10% depending on the channel. This day follows the money step by step: the card circuit and who earns what, the IBAN and the QR-bill, the agent network, four ways to send money to Dakar, a payment API in test mode — then a three-way negotiation, and your fintech's payment flow.

🕘 **09:00–12:15 · 13:30–16:45 · four 90-min blocks** | 💻 **laptop · no programming**

🎤 **the talk is short — 🛠️ most of the time, you do the work**

BY THE END OF DAY 2, YOU WILL BE ABLE TO

- Follow a card payment from authorisation to settlement, and split the merchant discount rate between issuer, scheme and acquirer.
- Check an IBAN, compute the check digit of a QR reference, and read the content of a QR-bill.
- Explain how electronic money is backed, and manage the liquidity of a mobile money agent.
- Compute the total cost of an international transfer — displayed fees *and* exchange-rate margin — across four real channels.
- Send requests to a payment API in test mode and read its responses, errors and webhooks.
- Describe your fintech's payment flow and revenue model.

BEFORE YOU START — TICK

- ☐ **Signed in** on this site (the button at the bottom right).
- ☐ **Day 1's work** at hand: your team's problem and customer (Day 1, section 1.9).

- [ ] **A spreadsheet open** for Block 2. Nothing to install: the API workshop in Block 3 runs on this page.

BLOCK 1 · 09:00–10:30 anatomy of a payment · interchange, IBAN, QR-bill

2.1 **Anatomy of a payment**
Four corners · authorisation, clearing, settlement · who earns what · 3-D Secure · instant payment and QR-bill
🎤 TALK · 20 MIN

2.2 **Interchange · IBAN · QR-bill**
The split simulator · checking an IBAN · the QR reference
🛠 WORKSHOP · 55 MIN

BLOCK 2 · 10:45–12:15 mobile money · Geneva → Dakar · the quiz

2.3 **Mobile money and transfers**
Electronic money · the agent and their two stocks · the true cost of a transfer
🎤 TALK · 15 MIN

2.4 **A day as an agent · Geneva → Dakar**
The agent simulator · CHF 200 through four channels, today's rates
🛠 WORKSHOP · 50 MIN

2.5 **End-of-morning quiz**
Eight questions · launched in class · counts toward participation
✅ QUIZ · 10 MIN

BLOCK 3 · 13:30–15:00 APIs and open banking · the console in test mode

2.6 **APIs and open banking**

Request and response · keys, tokens, idempotency, webhooks · PSD2, AISP and PISP

🎤 TALK · 20 MIN

2.7 **API workshop in test mode**

Eight missions in a payment console

🛠 WORKSHOP · 55 MIN

BLOCK 4 · 15:15–16:45 the three-way negotiation · your fintech's flow

2.8 **The three-way negotiation**

Merchant, provider, network · three confidential cards

🎤 TALK · 10 MIN

2.9 **The deal · your fintech's flow**

The deal calculator · the project's payment flow and revenue model

🛠 WORKSHOP · 65 MIN

← Course page

reset my Day 2 progress on this device

SECTION 2.1 · BLOCK 1

🎤 TALK · 20 MIN

Anatomy of a payment

Instructor's talk. If it does not take place, read this section on your own, then do the sorting exercise at the bottom of the page.

THE FOUR-CORNER MODEL

Five parties behind every card payment

Cardholder	Issuer	Scheme	Acquirer	Merchant
you, with the card	the bank that issued the card	Visa, Mastercard	the merchant's bank or PSP	the shop, the website

"Four corners": cardholder, issuer, acquirer, merchant. The scheme (card network) sits in the middle: it sets the rules, routes the messages and computes each party's position. Between the merchant and the acquirer there is often a **payment service provider** (Stripe, Adyen, Worldline…).

THE MECHANISM · FROM THE CARD TO THE MERCHANT'S ACCOUNT

Authorisation, clearing, settlement — eight steps

▸ A COFFEE PAID CHF 4.80 CONTACTLESS

1 · At the terminal. The card (or the phone) generates a single-use *cryptogram* from a secret key embedded in the chip: copying the number is not enough to clone a chip card. In Apple Pay or Google Pay, the real number is even replaced by a *token* specific to the phone.

EXAMPLE

Your phone touches the café's terminal. The real card number `4000 1234 5678 9010` does not leave it: the phone sends its token `4895 3700 0000 1102` and a cryptogram `8B2F61C4079E3A15`, computed from the amount, the date and a random number supplied by the terminal. Replayed tomorrow, that cryptogram would be refused.

2 · Authorisation request. The terminal sends the acquirer a message in **ISO 8583** format: amount (480 centimes), currency, card number or token, cryptogram, merchant identifier.

EXAMPLE

The message, field by field: `MTI 0100` (authorisation request) · `DE2 4895370000001102` (the token) · `DE3 000000` (a purchase) · `DE4 000000000480` (4.80) · `DE49 756` (Swiss franc) · `DE22 07` (contactless) · `DE42 CAFEDUCOIN0001` (the merchant) · `DE55` (the cryptogram).

3 · Routing. The acquirer forwards to the scheme (VisaNet, Banknet at Mastercard), which reads the first digits of the card — the **BIN**, 6 to 8 digits that identify the issuer — and routes the message to the right bank, anywhere in the world.

EXAMPLE

The network reads `489537`, the first six digits: they name the bank that issued the card. The token is first translated back to the real number by the network's token service, then the message goes to that bank. All of it takes a few tens of milliseconds.

4 · The issuer's decision. In a few milliseconds: valid card? correct cryptogram? sufficient balance or limit? And a **fraud-detection model** scores the transaction (place, time, amount, customer habits). Coded response: `00` approved, `51` insufficient funds, `05` declined.

EXAMPLE

At the issuer: card valid until 08/28 ✓ · cryptogram verified ✓ · balance CHF 1,250.00 ≥ 4.80 ✓ · fraud score 12 out of 999 (usual: café, neighbourhood, 8:15) ✓. Response `00`, authorisation code `A7K3Q9`. The same payment at 3 a.m. in Lagos, with the same card an hour after Geneva, would have scored high and got a `05`.

5 · Back to the terminal in one to two seconds: "Payment approved". No money has moved yet — the issuer has only *reserved* the amount on the cardholder's account.

EXAMPLE

The terminal receives `MTI 0110`, `DE39 00`: "Payment approved" appears. In your banking app: "Café du Coin · –4.80 · *pending*". Your available balance drops; your booked balance does not, yet.

6 · Clearing (in the evening). The merchant or its provider sends the batch of the day's transactions. The scheme computes, for each bank, what it owes and what it is owed — interchange and scheme fees included.

EXAMPLE

At 11 p.m. the café sends its batch: 312 payments, CHF 1,486.40. For your coffee the network records: amount 4.80 · interchange 0.44% = 0.02 owed to the issuer · scheme fee 0.15% = 0.01. The centimes of millions of payments become one net position per bank.

7 · Settlement (D+1). Net positions are settled between settlement banks: the issuer pays the acquirer the amount of the purchases *minus the interchange*.

EXAMPLE

The next day your bank pays: 4.80 – 0.02 interchange = 4.78 for this coffee, folded into one net transfer to the café's bank together with those of all its other customers. On your statement the line goes from "pending" to "booked".

8 · The merchant is credited by the acquirer, minus the MDR, one to three days later. For several weeks, the cardholder can still dispute: that is the *chargeback* window.

EXAMPLE

Two days later the café receives 4.80 – 1.5% commission = **CHF 4.73**. Of the 0.07 withheld: 0.02 for your bank, 0.01 for the network, 0.04 for the acquirer, which also pays for processing. If you disputed this coffee during the chargeback window (often 120 days), the café could have to give back the 4.80.

✓ All 8 steps are shown

WHO EARNS WHAT

The merchant discount rate, broken down

Merchant discount rate (MDR) = interchange + scheme fees + acquirer's margin

On a CHF 100 purchase with a 1.5% MDR, the merchant receives CHF 98.50. The **interchange** (set by the scheme, paid by the acquirer to the issuer) pays the issuer for the risk and credit it carries; the **scheme fees** pay Visa or Mastercard; the rest is the **acquirer's margin**, which also pays for the technical processing.

- **In the EU**, Regulation 2015/751 caps interchange on consumer cards at **0.2%** (debit) and **0.3%** (credit).
- **In Switzerland**, there is no law: agreements with the Competition Commission (COMCO) brought domestic credit-card interchange down to **0.44%** on average (since 2017).
- Interchange is what funds neobanks' "free" cards: they are issuers, and earn the interchange on every payment.

SECURITY AND FRAUD

Strong customer authentication, 3-D Secure, and who pays for fraud

The European **PSD2** directive requires **strong customer authentication** (SCA) for most online payments: two factors out of *knowledge* (a code), *possession* (the phone) and *inherence* (fingerprint, face). For cards, it goes through **3-D Secure**: the issuer asks for confirmation in its app. Version 2 is *risk-based*: the issuer receives around a hundred data points (device, history, amount) and lets low-risk payments through without friction.

The liability shift: when a payment has been authenticated with 3-D Secure, fraud is borne by the issuer. Without authentication, the cardholder disputes, the issuer raises a *chargeback* and the merchant loses the sale.

WITHOUT A CARD

Instant payment, TWINT, QR-bill

- **Instant payment.** In Switzerland, the SIC system (operated by SIX for the Swiss National Bank) has processed instant payments since August 2024: credited within seconds, 24 hours a day, 7 days a week, as ISO 20022 messages. In the EU, Regulation 2024/886 requires banks to receive and then to send SEPA instant transfers (ten seconds at most) at the price of an ordinary transfer, with verification of the payee's name.
- **TWINT.** The app is linked to the user's bank account; payment is made by scanning a QR code (or showing one), authorisation is given in real time, and the merchant is credited by its acquirer. No card network in the middle: TWINT is its own scheme.
- **The QR-bill.** Since 30 September 2022, it has replaced the orange and red payment slips. The QR code contains no money: it contains a *pre-filled payment order* — IBAN, amount, reference — which the payer's banking app reads and submits after confirmation.

CHECK · 3 MIN

Who does what in a card payment?

› CARDHOLDER · ISSUER · SCHEME · ACQUIRER · MERCHANT

Receives the interchange

Cardholder | Issuer | Scheme | Acquirer | Merchant

Pays the merchant discount rate (MDR)

Cardholder | Issuer | Scheme | Acquirer | Merchant

Routes the request to the right bank using the BIN

Cardholder | Issuer | Scheme | Acquirer | Merchant

Approves or declines the payment

Cardholder | Issuer | Scheme | Acquirer | Merchant

Credits the merchant, one to three days later

Cardholder | Issuer | Scheme | Acquirer | Merchant

Bears the fraud on a payment authenticated with 3-D Secure

Cardholder | Issuer | Scheme | Acquirer | Merchant

Sets the interchange schedule (under the authorities' oversight)

Cardholder | Issuer | Scheme | Acquirer | Merchant

0 of 7 sorted

SECTION 2.2 · BLOCK 1

WORKSHOP · 55 MIN

Interchange simulator · IBAN and QR-bill

A · 25 MIN · WHO WINS, WHO LOSES

Vary the MDR, the interchange, the fraud and the volume

An online merchant in Geneva. Each slider changes how the money is split between the four corners. Answer the three questions below the table, then explain what you observed.

Average basket (CHF) 50

Transactions per month 2,000

Merchant discount rate 1.50%

Interchange 0.44%

Scheme fees 0.15%

Acquirer's processing cost 0.25%

Fraud rate 0.10%

Without 3-D Secure | With 3-D Secure

PER MONTH · VOLUME CHF 100,000	CHF	PER CHF 100
Merchant — MDR + fraud (chargebacks)	-1,600.00	-1.60
Issuer — interchange	440.00	0.44
Scheme — fees	150.00	0.15
Acquirer — MDR – interchange – scheme fees – processing	660.00	0.66
Fraud lost, borne by the merchant	**100.00**	**0.10**

With the starting settings (MDR 1.5%, interchange 0.44%, scheme 0.15%, processing 0.25%), how much does the acquirer keep on a **CHF 100** purchase?

… CHF Check

Still the starting settings, without 3-D Secure: 2,000 transactions of CHF 50 per month and 0.10% fraud. Total payment cost for the merchant, **per month**?

… CHF Check

Interchange falls to **0.20%** (the EU debit cap) but the merchant discount rate stays at 1.5%. How much does the acquirer keep on CHF 100?

… CHF Check

✍️ **What you observed: which slider moves the most money, and towards whom? What does 3-D Secure change, and for whom?**

e.g. Lowering interchange to 0.2% increases the acquirer's margin if the MDR does not fall; with 3-D Secure, fraud shifts from the merchant to the issuer…

B · 20 MIN · THE IBAN AND THE QR-BILL

The check that prevents sending money to the wrong account

A Swiss IBAN has 21 characters: **CH**, two **check digits**, five **institution** digits (the bank), twelve **account** digits. To check it (ISO 13616 standard): move the first four characters to the end, replace each letter with a number (A = 10 … Z = 35; C = 12, H = 17), and divide the resulting large number by 97. **An IBAN is valid if the remainder is 1.** To *compute* the check digits, do the same with "00" in their place, and take 98 minus the remainder.

▸ IBAN CHECKER, STEP BY STEP

`CH93 0076 2011 6238 5295 7` Check

1. Move **CH93** to the end: `007620116238529 57CH93`
2. Letters become numbers: `007620116238529 57121793`
3. Remainder of division by 97: **1** → **valid IBAN**

Compute the two check digits of the Swiss IBAN whose remaining part is `0076 2011 6238 5295 7`: CH**??** 0076 2011 6238 5295 7. *(Tip: check "CH00 0076 2011 6238 5295 7" in the tool, then 98 – remainder.)*

… Check

A customer swapped the last two digits: `CH93 0076 2011 6238 5297 5`. What remainder do you get?

… Check

The **QR reference** (27 digits) ends with a check digit computed by “recursive modulo 10”: start with a carry of 0; for each digit, the new carry is entry (carry + digit) mod 10 of the table `0 9 4 6 8 2 7 1 3 5`; at the end, the check digit is (10 – carry) mod 10.

› QR REFERENCE CALCULATOR

`210000000003139471430000901` Compute

Successive carries: `0+2→4 · 4+1→2 · 2+0→4 · 4+0→8 · 8+0→3 · 3+0→6 · 6+0→7 · 7+0→1 · 1+0→9 · 9+0→5 · 5+0→2 · 2+3→2 · 2+1→6 · 6+3→5 · 5+9→8 · 8+4→4 · 4+7→9 · 9+1→0 · 0+4→8 · 8+3→9 · 9+0→5 · 5+0→2 · 2+0→4 · 4+9→6 · 6+0→7 · 7+1→3`

Check digit: **7** → full reference `210000000003139471430009017`

An SME numbers its invoices and derives the reference `000000000000002026102300 01` (26 digits) from them. What is the 27th, the check digit?

… Check

Here is the content of a QR-bill, as the code carries it (example from SIX’s implementation guidelines). One line per field; empty lines are optional fields left blank.

```
SPC   ← type
0200   ← version
1   ← coding
CH4431999123000889012   ← creditor IBAN
S   ← address type
Robert Schneider AG   ← name
Rue du Lac   ← street
1268   ← number
2501   ← postcode
Biel   ← town
CH   ← country
·   ← (ultimate creditor, 7 lines)
·
·
·
·
·
·
1949.75   ← amount
```

```
20  CHF   ← currency
21  S   ← debtor: address type
22  Pia-Maria Rutschmann-Schnyder   ← debtor: name
23  Grosse Marktgasse
24  28
25  9400
26  Rorschach
27  CH
28  QRR   ← reference type
29  210000000003139471430009017   ← reference
30  Order of 15 June 2020   ← message
31  EPD   ← end of data
```

TRUE OR FALSE?

Line 4: this is a **QR-IBAN**, because its institution number (31999) lies between 30000 and 31999.

True False

With a QR-IBAN, you can use a free reference in the international format (SCOR) instead of the QR reference.

True False

The last digit of the reference (line 29) is a recursive modulo 10 check digit.

True False

Scanning this code with your banking app sends the money immediately, without confirmation.

True False

The amount line can be left empty: the payer then enters the amount themselves.

True False

The QR-bill replaced the orange and red payment slips, which have no longer been accepted since 30 September 2022.

True False

0 of 6 sorted

DEBRIEF · 10 MIN

Two questions: why does interchange exist, and who should pay it? Why does a TWINT payment or a QR-bill cost the merchant less than a card payment?

SECTION 2.3 · BLOCK 2

🎤 TALK · 15 MIN

Mobile money and transfers

Instructor's talk, or self-study reading.

THE MECHANISM · ELECTRONIC MONEY

Where does the money in a Wave or Orange Money wallet come from?

- **One-for-one issuance.** When a customer hands 10,000 FCFA in cash to an agent, the **electronic money issuer** records 10,000 FCFA in their wallet. In return, the issuer holds 10,000 FCFA in a **safeguarding account** at a bank. The sum of the wallets always equals this account: electronic money is not created, it is *converted*.
- **The wallet is a claim** of the customer on the issuer, not a bank deposit: the issuer cannot lend this money. That is the BCEAO rule in the WAEMU and the rule of Directive 2009/110/EC in the EU.
- **The agent** is a neighbourhood shopkeeper who keeps two stocks: **cash** in the drawer and **electronic money** (the *float*) on the phone. A *deposit* (cash-in) means receiving cash and giving up float; a *withdrawal* (cash-out), the reverse. The agent earns a commission per transaction.
- **Rebalancing.** When the float or the cash runs out, the agent must go to the bank or to a super-agent to convert one into the other — a trip, time, a risk of theft. An agent who runs short turns customers away: that is the first breaking point of a network.

The measured effect. In Kenya, access to M-Pesa (launched in 2007) lifted about 2% of households — some 194,000 — out of extreme poverty, with a stronger effect for female-headed households (Suri and Jack, *Science*, 2016).

THE MECHANISM · AN INTERNATIONAL TRANSFER

What a transfer really costs: the fees *and* the rate

Total cost = amount sent − (amount received ÷ mid-market rate)

The **mid-market rate** is the midpoint between buy and sell on the foreign-exchange market. A provider advertising "zero fees" often makes its money on the rate. Between the

CFA franc and the euro, the rate is **fixed**: €1 = 655.957 FCFA; so CHF 1 = 655.957 ÷ (price of one euro in francs) FCFA.

- **The bank (SWIFT transfer)**: the money goes through a chain of correspondent banks, each of which may deduct fees; one to three business days.
- **Wise**: the money does not cross the border. Wise receives your francs in its Swiss account and pays the recipient from its local account; flows in both directions offset each other. Mid-market rate, displayed fees.
- **To a mobile wallet**: a transfer service credits the recipient's Wave or Orange Money wallet directly; the recipient cashes out at an agent or pays with it.
- **A stablecoin**: francs → USDC on an exchange (fees and spread), transfer on the blockchain (network fees), USDC → FCFA with a local intermediary (spread). Each conversion has its cost (Day 4).

United Nations Sustainable Development Goal 10.c: bring the cost of remittances below **3%** by 2030. The global average measured by the World Bank is about **6%** for a 200-dollar remittance, and sub-Saharan Africa is the most expensive region.

CHECK · 3 MIN

What happens to the agent's two stocks?

▸ CASH ↑ FLOAT ↓ · CASH ↓ FLOAT ↑ · NOTHING CHANGES

A customer deposits 10,000 FCFA to send to her mother

Cash ↑ float ↓ | Cash ↓ float ↑ | Nothing changes

A customer withdraws 25,000 FCFA received from his brother in France

Cash ↑ float ↓ | Cash ↓ float ↑ | Nothing changes

Two customers send each other money from phone to phone

Cash ↑ float ↓ | Cash ↓ float ↑ | Nothing changes

The agent goes to the bank to deposit surplus cash in exchange for float

Cash ↑ float ↓ | Cash ↓ float ↑ | Nothing changes

A customer pays the shopkeeper next door by scanning his QR code

Cash ↑ float ↓ | Cash ↓ float ↑ | Nothing changes

End of the month: a teacher withdraws the salary the State paid into his wallet

Cash ↑ float ↓ | Cash ↓ float ↑ | Nothing changes

0 of 6 sorted

SECTION 2.4 · BLOCK 2

🛠 WORKSHOP · 50 MIN

A day as an agent · CHF 200 from Geneva to Dakar

A · 20 MIN · THE AGENT NETWORK SIMULATOR

You run a Wave point in Dakar. Sixteen customers today.

You start with **150,000 FCFA** in cash and **150,000 FCFA** of float. Each transaction served earns you **1%**. Before each customer, you can **rebalance** (go to the super-agent and split half-and-half): **500 FCFA** in transport. A customer you cannot serve goes to the competitor: **–2,000 FCFA** (the commissions they will no longer bring you). The maximum possible is 4,950 FCFA.

CASH	FLOAT (ELECTRONIC MONEY)	TODAY'S EARNINGS
150,000	150,000	0
		0 CUSTOMER(S) LOST

Customer 1 of 16 · **Awa** wants to **deposit 20,000 FCFA**.

Serve the customer | Rebalance first (–500)

The day's log appears here.

✍ **Your score, and your strategy in one sentence: when did you rebalance, and why at that moment?**

The score is inserted at the end of the game; add your strategy.

B · 30 MIN · WITH TODAY'S REAL RATES

Sending CHF 200 from Geneva to Dakar through four channels

In pairs. For each channel, go to the provider's website or simulator *today*, simulate sending CHF 200 to Senegal, and note the fees, the exchange rate offered and the

announced delivery time. First enter today's euro rate in francs (from the SNB, Wise or your bank's website): the table derives the CHF → FCFA mid-market rate from it.

Price of one euro in francs today → **mid-market rate: CHF 1 = 705.33 FCFA**

0.93

CHANNEL	FEES (CHF)	RATE OFFERED (FCFA PER CHF 1)	DELIVERY TIME	RECEIVED (FCFA)	TOTAL COST (CHF)	COST %
Bank · international transfer (SWIFT)						
Wise						
To a Wave wallet						
Stablecoin (CHF → USDC → FCFA)						

Copy the table into my notebook ↓ | Fill with example values

The example values are fictitious; they only show how the table calculates. The task is to collect the real ones.

✍️ **The table, then your conclusion: which channel for which customer (who has an account, a smartphone, needs cash the same day)?**

SECTION 2.5 · BLOCK 2

✅ QUIZ · 10 MIN

End-of-morning quiz

INTERACTIVE QUIZ · COUNTS TOWARD PARTICIPATION

Eight questions, one attempt each, all explained

Launched in class; this page unlocks by itself. On your own, without an AI assistant. A missed quiz scores zero, unless arranged in advance.

🔒 **The quiz starts together.** It is launched in class; this page unlocks by itself within seconds, without reloading.

0 / 8 answered

1. **In the four-corner model, interchange is paid…**

- by the merchant to the scheme
- by the acquirer to the issuer
- by the issuer to the acquirer
- by the cardholder to their bank

2. **The merchant discount rate (MDR) breaks down into…**

- cardholder fees + tax
- interchange only
- issuer’s margin + VAT
- interchange + scheme fees + acquirer’s margin

3. **PSD2 strong customer authentication requires…**

- two factors out of knowledge, possession and inherence
- a twelve-character password
- a handwritten signature
- physical presence at a branch

4. An instant payment (SIC in Switzerland, SEPA Instant in the EU) is credited…

- the next business day
- at the end of the day, in batches
- within seconds, 24 hours a day, 7 days a week
- after manual approval by the bank

5. The funds backing an electronic money issuer’s electronic money are…

- deposited one-for-one in a safeguarding account at a bank
- lent to SMEs to finance the network
- invested in shares
- non-existent: electronic money is created by the operator

6. For a mobile money agent, a customer’s withdrawal (cash-out)…

- increases their cash and decreases their float
- does not change their stocks
- decreases both their cash and their float
- decreases their cash and increases their float

7. **In a payment API, a CHF 49.90 purchase is sent with…**

- "amount": 49.90
- "amount": 4990
- "amount": "CHF 49.90"
- "amount": 49

8. **An app that triggers a transfer from your bank account, with your consent, to pay for an online purchase is, in PSD2 terms…**

- an AISP
- a card issuer
- a PISP
- an acquirer

SECTION 2.6 · BLOCK 3

🎤 TALK · 20 MIN

APIs and open banking

Instructor's talk, or self-study reading.

WHAT AN API IS

A request, a response — like a form filled in by a program

An **API** (application programming interface) lets one program call another. Payment APIs are almost all **REST**: the program sends an HTTP *request* to an address (the *endpoint*) and receives a *response* in JSON.

```
POST https://api.stripe.com/v1/payment_intents          ←   method + endpoint
Authorization: Bearer sk_test_…                         ←   the secret key
Idempotency-Key: order-2026-1043                        ←   duplicate guard
{ "amount": 4990, "currency": "chf" }                       ←   the body

200 OK
{ "id": "pi_3Q…", "status": "requires_payment_method", "amount": 4990 }
```

- **Methods**: `GET` reads, `POST` creates or triggers, `DELETE` deletes.
- **Status codes**: `200` success · `400` malformed request · `401` not authenticated · `402` payment declined · `404` not found · `5xx` server-side failure.
- **Amounts in the smallest unit**: CHF 49.90 is written `4990` (centimes) — except for currencies without decimals, such as the CFA franc: 10,000 FCFA is written `10000`.

WHAT PROTECTS

Keys, tokens, idempotency, webhooks

- **Keys.** The *secret key* (`sk_test_…` in test, `sk_live_…` in production) gives full rights over the account: it stays on the merchant's server, never in a web page or an app. The *publishable key* (`pk_…`) may be in the page; it only allows card data to be transmitted.
- **Tokens.** The card number goes directly from the browser to the provider, which returns a *token* (`pm_…`) to the merchant. The merchant never sees the card: it falls almost entirely outside the scope of the PCI DSS security standard.
- **Idempotency.** If the connection drops during a payment, the program resends the request. With the same `Idempotency-Key`, the provider returns the result of the first

one instead of charging twice.

- **Webhooks.** Some results arrive later (3-D Secure confirmation, bank transfer). Instead of asking again and again, the merchant provides an address; the provider calls it when the status changes (`payment_intent.succeeded`). The call is signed, so that an attacker cannot announce a fake payment.

OPEN BANKING

PSD2: banks must open accounts to authorised third parties

Since PSD2 (Directive 2015/2366, applied since 2018), a European bank must give access to payment accounts, *with the customer's consent*, to two types of licensed providers:

- **AISP** — account *information* service provider: it *reads* balances and transactions (budget aggregator, accounting tool, credit scoring).
- **PISP** — payment *initiation* service provider: it *triggers* a transfer from the customer's account (paying online by bank transfer, without a card).

The customer gives consent in their bank's app (OAuth 2.0 flow, with strong customer authentication); the third party receives an *access token* limited in time and in rights, never the password. The APIs follow standards such as the Berlin Group's NextGenPSD2. **Switzerland has no legal obligation**: opening up happens through market agreements, notably via SIX's bLink platform. In the EU, the revision (PSD3 and the Payment Services Regulation) is in the process of adoption.

CHECK · 3 MIN

AISP, PISP, or neither?

› CLASSIFY EACH SERVICE

A budgeting app that shows your accounts at three banks together

AISP PISP Neither

A "pay by bank transfer" button on a merchant website that debits your account after confirmation in your bank's app

AISP PISP Neither

An online lender that analyses your last twelve months of transactions to decide on a loan

AISP PISP Neither

A neobank that issues you a debit card

AISP PISP Neither

Accounting software that executes an SME’s supplier payments from its bank account

AISP PISP Neither

Paying in a shop with a Visa card

AISP PISP Neither

0 of 6 sorted

SECTION 2.7 · BLOCK 3

🛠 WORKSHOP · 55 MIN

API workshop · a payment platform in test mode

IN TEAMS · EIGHT MISSIONS

Create a customer, collect CHF 49.90, get declined, pass 3-D Secure, refund

The console below imitates Stripe's API in test mode, simplified: JSON body, fictitious ids, no real money, nothing leaves your browser. Choose an endpoint, click "Template" to pre-fill the body, edit it, send, read the response and the webhook log. Confirm and retrieve act on the last payment created (or on the one whose `"id"` you put in the body). The missions tick themselves off.

1. Create a customer (`POST /v1/customers`)
2. Create a payment of **CHF 49.90** for this customer — mind the unit
3. Confirm this payment with `pm_card_visa` and receive the success webhook
4. Create another payment and get it **declined** for insufficient funds — read the `decline_code`
5. Create a payment that requires **3-D Secure**, then have it confirmed
6. Send the same payment creation twice with the same `Idempotency-Key` — only one payment must exist
7. Trigger a **401** error by replacing the secret key with a publishable key `pk_test_…`
8. Refund the successful payment (`POST /v1/refunds`)

REQUEST

POST /v1/customers

Authorization: Bearer sk_test_mfb545_demo

Idempotency-Key: (optional)

```
{
  "name": "Awa Diop",
  "email": "awa.diop@example.com"
}
```

Template Send →

Test payment methods: `pm_card_visa` (accepted) · `pm_card_chargeDeclinedInsufficientFunds` (declined, insufficient funds) · `pm_card_authenticationRequired` (3-D Secure required). These are the ones from Stripe's documentation; with a real Stripe account in test mode (free), the same requests work.

RESPONSE

```
_
```

WEBHOOKS RECEIVED BY YOUR SERVER

none yet

✍️ **Two short answers: why does the merchant wait for the webhook before shipping the order? What could someone do who finds your secret key in your website's code?**

e.g. The final status may arrive after 3-D Secure…; with the secret key one can issue refunds, read the customers…

DEBRIEF · 15 MIN

Which mission caused trouble, and which error message put you on the right track? Where, in your own fintech's customer journey, will a third-party API do part of the work?

SECTION 2.8 · BLOCK 4

🎤 INSTRUCTIONS · 10 MIN

Role play · the three-way negotiation

THE SITUATION

An online merchant, its payment provider and a card network

Teranga Market, a Geneva online shop (groceries and cosmetics for the West African diaspora), collects **CHF 1.2 million** a year by card. Its provider, **PayLéman**, charges it 1.8%. A competitor offers 1.4%. Online fraud reaches 0.4% of sales, above the monitoring threshold of the card network **Kora**. The three parties meet to set: **the MDR**, **who pays for fraud**, and **whether 3-D Secure is activated**.

The rules. The three project teams form three tables; at each table, one person per role (three per table). Read *only your own card*. 20 minutes of negotiation; a deal is recorded in the calculator in the next section. No deal is a possible outcome — to be compared with your fallback option.

🛒 The merchant

Teranga Market

🏦 The provider

PayLéman SA, acquirer

🌐 The network

Kora, card network

SECTION 2.9 · BLOCK 4

🛠 NEGOTIATION AND PROJECT · 65 MIN

The deal, then your fintech's payment flow

20 MIN · NEGOTIATION · 15 MIN · DEBRIEFING

Enter the terms of your deal: the calculator gives each party's result

⏱ Negotiation **19:56** Pause ↺

Agreed MDR 1.80%

Share of remaining fraud paid by the merchant 100%

3-D Secure activated

No

Network rebate (0.05%) if 3-D Secure

No

3-D Secure integration paid by

the merchant

Copy the terms into my notebook ↓

PER YEAR	RESULT	OBJECTIVE / FALLBACK	
🛒 Merchant — total cost (MDR + fraud + lost sales)	CHF 26,400	objective < 21,000 · fallback 24,600	✗ worse than the fallback
🏦 Provider — profit	CHF 10,200	objective ≥ 4,800 · fallback 0	✓ objective
🌐 Network — net fees	CHF 1,200	objective: fraud < 0.3%	fraud 0.40%

Without 3-D Secure: fraud 0.40% (CHF 4,800), network penalty of CHF 1,200 for the provider. As soon as you change a term, the deal is recorded.

✍ **Your role, the terms of the deal (or the lack of one), and the argument that moved the table.**

The calculator's button inserts the terms; add your role and the decisive argument.

At the debriefing, the three tables compare their deals. The question to ask: *was there a deal better for all three*? What did one need to know that only one role knew?

30 MIN · IN TEAMS · THE PROJECT

The payment flow and revenue model of your pension-savings app

Both teams' project: a pension-savings app for Senegal, Congo or a neighbouring country, with small monthly payments invested in Swiss blue-chip equity funds until 65. First draw the flow on paper — from the customer's mobile wallet to the fund units in Switzerland, then the way back at 65: who pays whom, by what means, how long it takes, who holds the money in the meantime — then describe it here. Each member fills in their notebook — the same version for the whole team.

1. The flow: who pays whom, on which rail (card, instant payment, TWINT, mobile money, stablecoin…), how fast, who holds the money in the meantime, and under which licence.

e.g. The customer pays 10,000 FCFA from Orange Money → collection account at a partner bank in Dakar (D+0) → monthly pooling, FCFA → CHF conversion → custodian bank in Switzerland → fund units bought in each customer's name (D+…); at 65: units sold → CHF → FCFA → mobile wallet…

2. The revenue model: who pays, the mechanism (a % of assets, a fee per payment, FX margin, subscription…), the price, and an estimated ARPU with your calculation. For an app that must earn trust, also say how each fee is shown to the customer.

e.g. 0.8% a year of assets, no hidden FX margin; average balance 300,000 FCFA → ARPU 200 FCFA a month; variable cost: …

BEFORE DAY 3 · SATURDAY 24 OCTOBER (THE NEXT DAY)

- Check that your Google account opens Google Colab ↗: tomorrow's Block 1 is a guided credit-scoring notebook.
- Day 3 adds the pension-savings plan itself: note how much your customer could pay in each month, from what age, and what would make them trust you — or run.

DAY 3 · SATURDAY 24 OCTOBER 2026 · MFB 545

Digital credit, neobanks and financial inclusion

A loan granted in thirty seconds in an app rests on a model that turns data into a probability of default, and on a threshold someone chose. A neobank with fifty million customers can earn more than a billion; another, with more revenue per customer, loses money. This day opens both black boxes: you build a score on 1,000 real loans, set the threshold, redo a neobank's accounts, lend to strangers on a simulated platform, debate buy now, pay later — then design your app's pension-savings plan.

🕘 **09:00–12:15 · 13:30–16:45 · four 90-min blocks** 💻 **laptop · a Google account for Colab**

🎤 **the talk is short —** 🛠️ **most of the time, you are the ones working**

BY THE END OF DAY 3, YOU WILL BE ABLE TO

- Explain how a credit score turns data into a probability of default, and read an AUC and a Gini.
- Choose an acceptance threshold by trading off default rate, volume and result — and calculate an expected loss.
- Calculate a neobank's ARPU, customer lifetime value and acquisition-cost payback period.
- Bring a neobank's published income statement down to one customer, and say why some lose money.
- Tell the forms of crowdfunding apart, measure the risk of a peer-to-peer loan portfolio, and argue the status of buy now, pay later.
- Design your app's pension-savings plan: payments, fees, capital at 65, transparency and safeguards.

BEFORE YOU START — TICK

- ☐ **Signed in** on this site (the button at the bottom right).

- [ ] **Signed in to Google** in the same browser: the Block 1 notebook opens in Google Colab ↗. Without an account, the simulator in section 3.2 is enough.
- [ ] **Your work from Days 1 and 2** at hand: your team's problem, customer, payment flow and revenue model (Day 2, section 2.9).

BLOCK 1 · 09H00–10H30 how a score works · the notebook · threshold versus profitability

3.1

How a credit score works

Expected loss · from data to decision · AUC and Gini · the threshold · KKG, Directive 2023/2225, AI Act, usury rate

🎤 TALK · 20 MIN

3.2

Live scoring · threshold versus profitability

The Colab notebook on 1,000 loans · 300 test loans and the threshold · one applicant and their two reasons

🛠 WORKSHOP · 55 MIN

BLOCK 2 · 10H45–12H15 neobank economics · CAC, LTV, churn · the quiz

3.3

The economics of a neobank

Five revenue lines · ARPU, LTV, CAC, churn · Revolut, N26, neon, Yuh

🎤 TALK · 15 MIN

3.4

CAC, LTV · the income statement per customer

The unit-economics simulator · rebuilding published accounts

🛠 WORKSHOP · 50 MIN

3.5

End-of-morning quiz

Eight questions · launched in class · counts toward participation

✅ QUIZ · 10 MIN

BLOCK 3 · 13H30–15H00 crowdfunding, peer-to-peer lending · the BNPL debate

3.6 **Crowdfunding, peer-to-peer lending, BNPL**

Four forms · a P2P loan step by step · buy now pay later and its status

🎤 TALK · 20 MIN

3.7 **A campaign, a portfolio, a debate**

What is left of a campaign · 2,000 simulated years of loans · “is BNPL credit?”

🛠 WORKSHOP · 55 MIN

BLOCK 4 · 15H15–16H45 le crédit digital en Afrique · le plan d’épargne-retraite

3.8 **Digital credit in Africa**

M-Shwari and what we know about it · M-Baobab · the abuses and Kenya’s response

🎤 TALK · 15 MIN

3.9 **Your app’s pension-savings plan**

The plan at 65, fees, transparency, safeguards · peer review

🛠 WORKSHOP · 75 MIN

← Course page

reset my Day 3 progress on this device

SECTION 3.1 · BLOCK 1

🎤 TALK · 20 MIN

How a credit score works

Instructor's talk. If it does not take place, read this section on your own, then do the sorting exercise at the bottom of the page.

THE LENDER'S QUESTION

How much will this loan cost me, on average?

Expected loss (EL) = PD × LGD × EAD

- **PD** — *probability of default*: the probability that the borrower stops paying (usually: more than 90 days late) over the life of the loan or over one year. This is what the score estimates.
- **LGD** — *loss given default*: the share of the amount the lender does not get back, after collection. For unsecured consumer credit, often 40 to 70%.
- **EAD** — *exposure at default*: the amount owed at the moment of default.

A CHF 5,000 loan, PD 12%, LGD 50%: expected loss 0.12 × 0.50 × 5,000 = **CHF 300**. The price of credit must cover this loss, the cost of money, servicing costs and the capital tied up. This is **risk-based pricing**: two customers of the same product can pay two different rates.

THE MECHANISM · FROM DATA TO DECISION

Seven steps between the application and the app's "yes"

▸ AN APPLICATION FOR A CHF 3,000 LOAN OVER 24 MONTHS

1 · The application. The customer fills in the app: amount, term, income, expenses, job. Identity is checked by the Day 1 eKYC chain.

2 · External data, with consent. The credit register (ZEK and IKO in Switzerland, a credit information bureau in WAEMU) gives current loans and arrears. If the customer allows it, an **open-banking API** (Day 2, an AISP) reads twelve months of statements; in Africa, the mobile-wallet history plays that role.

3 · The variables. Raw data is turned into a few dozen variables: “negative average balance”, “number of payments more than 30 days late”, “salary paid in regularly”, “expenses-to-income ratio”. This is the step that makes a score good or bad.

4 · The model. A **logistic regression** adds up the weighted variables — the *logit* — then turns it into a probability: PD = $1 \div (1 + e^{-\text{logit}})$. Banks often translate it into **points** (a scorecard where, for example, +20 points double the good-payer odds). Digital lenders also use boosted decision trees, more accurate and harder to explain.

5 · Rules before the score. Some refusals never reach the model: minimum age, residence, detected fraud, an IKO entry, insufficient credit capacity (in Switzerland, the KKG requires that the credit can be repaid within 36 months without touching the subsistence minimum).

6 · Threshold and price. PD below the threshold: accepted, and the rate is set according to the PD (risk-based pricing), within the legal cap. Above: refused — or a smaller amount.

7 · Afterwards. The loan is monitored: every late payment is reported to the register, every repayment feeds the next score. The model itself is revalidated every year on recent loans, because the economy changes and the score degrades.

✓ All 7 steps are shown

MEASURING A SCORE

AUC and Gini: does the score rank the good before the bad?

A score does not say who will default; it *ranks*. It is judged on loans it has never seen. The **AUC** is the probability that a randomly drawn bad payer gets a higher PD than a randomly drawn good payer: **0.5** = a coin toss, **1** = perfect ranking. Banks rather report the **Gini = 2 × AUC – 1**. A consumer-credit score typically has a Gini between 0.5 and 0.7; above 0.9 on test data, suspect a mistake (a variable that “knows” the answer).

The threshold. The model gives a PD; a person sets the *threshold* above which applications are refused. If a repaid loan earns a margin m (as a % of the amount) and a default costs the LGD, a loan is worth making as long as $(1 - \text{PD}) \times m > \text{PD} \times \text{LGD}$, that is:

Theoretical threshold: PD < m ÷ (m + LGD)

With a 15% margin and a 50% LGD: 0.15 ÷ 0.65 = **23.1%**. Accepting more widely raises volume and defaults; accepting more tightly turns away good customers. The simulator in the next section shows the trade-off on real loans.

THE FRAMEWORK

What the law requires of a score, in Switzerland, the EU and WAEMU

- **Switzerland.** The Consumer Credit Act (KKG/LCC) requires a **creditworthiness assessment** and a check with the Consumer Credit Information Centre (**IKO**); the **ZEK** is the lenders' private credit register. Maximum annual effective rate in 2026: **10%** for a cash loan, **12%** for an overdraft or a credit card.
- **European Union.** The new Consumer Credit Directive (2023/2225, applicable from **20 November 2026**) requires a creditworthiness assessment based on relevant data, forbids using sensitive data (health, origin…) and social-media data in it, and gives a customer refused by an automated decision the right to an **explanation** and to **human intervention**. The **AI Act** classes credit scoring of individuals as a **high-risk** AI system. And the Court of Justice ruled in 2023 (SCHUFA, C-634/21) that the score itself can be an automated decision under the GDPR when the lender relies on it decisively.
- **WAEMU.** The **usury rate** set by the WAMU Council of Ministers caps the cost of credit: **14%** a year for banks since 1 June 2026 (15% before), **24%** for microfinance institutions and financial institutions. Credit information bureaus (BIC) licensed by the BCEAO share repayment histories.

CHECK · 4 MIN

Can this data go into a score?

▸ USABLE · SENSITIVE, MUST BE JUSTIFIED · EXCLUDE

Late payments over the last two years, from the credit register

Usable | Sensitive, must be justified | Exclude

The applicant's sex

Usable | Sensitive, must be justified | Exclude

Age

Usable | Sensitive, must be justified | Exclude

Twelve months of bank statements read through an open-banking API, with the customer’s consent

Usable | Sensitive, must be justified | Exclude

The home postcode

Usable | Sensitive, must be justified | Exclude

The phone’s contact list and social-media activity

Usable | Sensitive, must be justified | Exclude

The history of deposits, transfers and payments in a mobile-money wallet, with consent

Usable | Sensitive, must be justified | Exclude

Nationality

Usable | Sensitive, must be justified | Exclude

0 of 8 sorted

SECTION 3.2 · BLOCK 1

🛠 WORKSHOP · 55 MIN

Live scoring · threshold versus profitability

A · 25 MIN · THE NOTEBOOK

Build the score on 1,000 real loans

The public “German Credit” dataset (UCI, 1,000 consumer loans, 300 of which went bad) is the classic training set for scoring. The notebook loads it, looks at the default rate by category, fits a logistic regression on seven variables, measures the AUC and computes the portfolio result threshold by threshold. Run the cells in order (▶ or `Shift + Enter`); there is no line to write.

Open the notebook in Colab ↗ | Download the .ipynb

If Colab asks for confirmation (“This notebook was not authored by Google”), choose “Run anyway”. Without a Google account: do part B, which uses exactly the same model and the same 300 test loans.

Cell 5 of the notebook: what is the **AUC on the test sample**, rounded to two decimals?

… Check

With a 15% margin and a 50% LGD, what is the **theoretical threshold**, as a % PD?

… % Check

✍ **The variable with the highest factor on the odds, and what it means in one sentence. What you saw in cell 7 (sex, nationality), and whether you would use those variables.**

e.g. A poor credit history multiplies the odds of default by 3.1… ; foreign workers have a higher default rate here, but…

B · 20 MIN · THE “THRESHOLD VERSUS PROFITABILITY” SIMULATOR

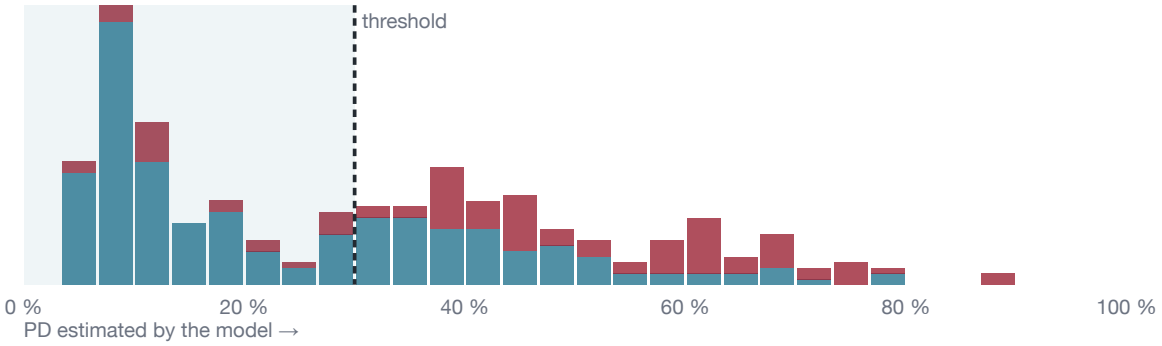
The 300 test loans, and the threshold in your hands

Each bar counts some of the 300 loans the model has not seen, placed by their PD: **blue**, repaid; **red**, defaulted. Everything to the left of the threshold is accepted. Amounts from the dataset, read as francs for the exercise.

Threshold: accept if the PD is below 30 %

Margin on a repaid loan 15 %

Loss given default (LGD) 50 %



152 / 300	13.8 %	76	72 / 93
ACCEPTED	DEFAULTS AMONG ACCEPTED	GOOD CUSTOMERS REFUSED	DEFAULTS AVOIDED

15,831
PORTFOLIO RESULT

Find the best threshold | Copy my choice into my notebook ↓

Accept everyone: 300 loans, 93 defaults (31%), result -99,825. Theoretical threshold with these assumptions: 23.1 %.

Set the margin back to 15% and the LGD to 50%, and put the threshold at **20%**. How many loans out of 300 are accepted?

… loans Check

C · 10 MIN · ONE APPLICANT, ONE DECISION, TWO REASONS

The same model, applied to one person

The notebook's model, with its coefficients. Change the profile: the PD, the decision at the threshold chosen in B, and the two reasons that weigh most. This is what a bank must be

able to tell a refused customer.

Current account

negative balance

Term 24 months

Amount 3,000

Savings below 100

yes

Age 30 years

Poor credit history

no

49.4 %	488	refused
ESTIMATED PD	SCORE IN POINTS	AT THE 30 % THRESHOLD (SECTION B)

The two reasons that raise the risk most, compared with the average applicant: the current account (negative balance) (+0.80 on the logit) ; savings below 100 (+0.18 on the logit). Score in points: 600 for odds of 50 good to 1 bad, +20 points each time the odds double.

An applicant: current account with a negative balance, 24 months, 3,000, savings below 100, aged 30, normal history. What is their **PD**, in %?

… % Check

✍️ **Your threshold, what it gives (accepted, defaults, result), and what you would say to a customer refused at a 30% PD.**

The simulator's button inserts the threshold and its figures; add your reasoning.

DEBRIEF · 15 MIN

Three questions: which threshold did you choose, and what moved it — the margin, the LGD, fear of defaults? How many good customers does the best threshold turn away? Is the score "fair" if it refuses more often a group that defaults more often?

SECTION 3.3 · BLOCK 2

🎤 TALK · 15 MIN

The economics of a neobank

Instructor's talk, or self-study reading.

WHERE THE MONEY COMES FROM

Five revenue lines, all small per customer

- **Interchange** (Day 2): the neobank is the issuer; it earns a share of every card payment. In the EU, capped at 0.2% on debit: CHF 800 of monthly spending brings in CHF 1.60.
- **FX**: a margin on the rate, often zero on weekdays up to a cap, then marked up (at weekends, above the limit, on illiquid currencies).
- **Subscriptions**: a few percent of customers pay for a plan (travel insurance, more free FX, a metal card).
- **Interest margin**: deposits placed with the central bank or in government securities earn interest; the neobank passes part of it on. With rates at zero (Switzerland 2015–2022), this line disappears.
- **Credit and investing**: overdrafts, loans, buy now pay later, brokerage. The most profitable lines — and the ones that need a banking licence and a score.

THE MECHANISM · UNIT ECONOMICS

What a customer is worth, and what they cost to acquire

ARPU = average revenue per active customer per month

Customer lifetime value (LTV) = ARPU × contribution margin ÷ monthly churn

Payback period = CAC ÷ (ARPU × contribution margin)

The **contribution margin** is what remains of ARPU after variable costs (card processing, support, fraud, compliance per customer). **Churn** is the share of customers who leave — or go inactive — each month: at 2% a month, a customer stays 1 ÷ 0.02 = 50 months on average. **CAC** is the cost of acquiring a customer (advertising, referral bonuses, onboarding

and identity-verification costs). The industry rule: **LTV ≥ 3 × CAC**, and CAC recovered in under 12 to 18 months.

FOUR CASES

Revolut, N26, neon, Yuh: why some lose money with millions of customers

- **Revolut** (London, 2015): 52.5 million customers at the end of 2024, revenue £3.1 billion, profit before tax £1.1 billion — a fourth profitable year. The engine: many products per customer (FX, subscriptions, credit, trading, interest) and global scale on a single platform.
- **N26** (Berlin, 2013): 4.2 million "revenue-relevant" customers at the end of 2023, revenue €313.5 million, net loss €102.4 million. Growth held back from 2021 to 2024 by BaFin, which had capped new account openings over anti-money-laundering weaknesses.
- **neon** (Zurich, 2019): about 250,000 customers at the end of 2025; no banking licence of its own (accounts are held by Hypothekarbank Lenzburg), a low-fee, subscription model.
- **Yuh** (2021, Swissquote and PostFinance; 100% Swissquote since July 2025): profitable at the end of 2024, just under 400,000 accounts at the end of 2025. Revenue comes largely from investing and the interest margin rather than from cards.

Sources: the companies' annual reports and press releases (Revolut, Annual Report 2024; N26, press release of 13 Nov. 2024; Swissquote, Annual Report 2025; PostFinance, "Yuh 2024" release). The figures change every year: the workshop asks you to look up the latest published ones.

CHECK · 3 MIN

Where does the neobank's revenue come from, in each case?

▸ INTERCHANGE · FX · SUBSCRIPTION · INTEREST MARGIN · CREDIT

You pay CHF 4.80 for a coffee in Lausanne with your neobank card

Interchange | FX | Subscription | Interest margin | Credit

CHF 2,000 stays in your account all month

Interchange FX Subscription Interest margin Credit

You switch to the Premium plan at 9.99 a month for the travel insurance

Interchange FX Subscription Interest margin Credit

On a Saturday, you change CHF 500 into CFA francs in the app

Interchange FX Subscription Interest margin Credit

You spread the purchase of a laptop over six months in the app

Interchange FX Subscription Interest margin Credit

You pay €120 in a restaurant in Paris with the same card

Interchange FX Subscription Interest margin Credit

0 of 6 sorted

SECTION 3.4 · BLOCK 2

🛠 WORKSHOP · 50 MIN

CAC, LTV, churn · a neobank's income statement

A · 25 MIN · THE UNIT-ECONOMICS SIMULATOR

Build ARPU line by line, then see whether the customer is worth what they cost

A fictitious neobank. Start from one of the three profiles, then move the sliders. Find **the slider that pushes LTV below CAC** the fastest.

EU neobank, card only | Swiss neobank | Multi-product

Card spending per month 800 CHF

Interchange 0.20 %

Share of customers on a paid plan 10 %

Plan price per month 9.00 CHF

Average deposit 2,000 CHF

Annual interest margin on deposits 1.50 %

FX per month 150 CHF

FX margin 0.50 %

Variable cost per customer per month 2.50 CHF

Monthly churn 1.50 %

Acquisition cost (CAC) 80 CHF

PER ACTIVE CUSTOMER PER MONTH	CHF	SHARE OF ARPU
Interchange	1.60	28 %
Subscriptions	0.90	16 %
Interest margin	2.50	43 %
FX	0.75	13 %
ARPU	**5.75**	**100 %**

– variable cost	-2.50	
Contribution	**3.25**	

67 months	217	2.71	24.6 months
AVERAGE LIFETIME	LTV (CHF)	LTV ÷ CAC (TARGET ≥ 3)	PAYBACK PERIOD

Illustrative profiles, built to show the mechanisms: these are not the figures of any real bank.

A neobank: ARPU CHF 6 a month, contribution margin 60%, churn 2% a month. What is the **LTV**?

… CHF Check

The same, but churn rises to **4%** a month (customers open the account for the bonus, then abandon it). LTV?

… CHF Check

With a CAC of CHF 90 and the same contribution of CHF 3.60 a month, what is the **payback period**?

… months Check

✍ **Which slider pushes LTV below CAC the fastest? What would a neobank actually do to fix it?**

e.g. Churn: from 1.5% to 3% a month, LTV is halved… ; the fix: make the account the main account (salary paid in)…

B · 25 MIN · IN PAIRS · THE INCOME STATEMENT PER CUSTOMER

Rebuild a neobank's accounts from its published figures

Two columns are prefilled with published figures. In the third, put a neobank of your choice — neon, Yuh, Wise, Monzo, Nubank, or an African neobank (Kuda, TymeBank,

Moniepoint…) — using **its latest annual report or press release**, found online now. The table brings everything down to one customer.

	REVOLUT · 2024	N26 · 2023	your neobank · year
Currency	£	€	CHF
Customers (millions)	52.5	4.2 "revenue-relevant"	
Revenue (millions)	3,100	313.5	
Result (millions)	1,100 before tax	–102.4 net	
Revenue per customer	**£ 59.05**	**€ 74.64**	
Costs per customer (revenue – result)	**£ 38.10**	**€ 99.02**	
Result per customer	**£ 20.95**	**€ -24.38**	
Source	Annual Report 2024	press release, 13.11.2024	link or document

Copy the table into my notebook ↓

Revolut 2024: what is the **revenue per customer**, in pounds?

… £ Check

N26 2023: what is the **net result per customer**, in euros?

… € Check

✍️ **The table, your source, then: N26 has more revenue per customer than Revolut and loses money. Why? And your neobank?**

SECTION 3.5 · BLOCK 2

✅ QUIZ · 10 MIN

End-of-morning quiz

INTERACTIVE QUIZ · COUNTS TOWARD PARTICIPATION

Eight questions, one attempt each, all explained

Launched in class; this page unlocks by itself. On your own, without an AI assistant. A missed quiz scores zero unless arranged in advance.

🔒 **The quiz starts together.** It is launched in class; this page unlocks by itself within seconds, without reloading.

0 / 8 answered

1. **A CHF 10,000 loan, PD 5%, LGD 60%. The expected loss is…**

- CHF 500
- CHF 6,000
- CHF 300
- CHF 30

2. **A credit score has an AUC of 0.5 on the test sample. This means…**

- it ranks no better than chance
- it is wrong about defaults half the time
- it accepts 50% of applications
- it is excellent: 0.5 is the norm

3. **The lender raises its acceptance threshold (it accepts higher PDs). In general…**

volume falls and the default rate falls

nothing changes if the model is good

volume falls and the default rate rises

volume rises and the default rate among accepted loans rises

4. A 15% margin on a repaid loan, a 50% LGD. The theoretical threshold is about…

15 %

23 %

50 %

65 %

5. A neobank's customer lifetime value (LTV) is calculated as…

CAC × churn

ARPU × contribution margin ÷ monthly churn

ARPU × number of customers

total revenue ÷ CAC

6. Monthly churn rises from 2% to 4%. LTV…

rises by 2%

does not change

doubles

is halved

7. Why does a European neobank earn little on each debit-card payment?

- Because interchange is capped at 0.2% in the EU
- Because Visa keeps the whole fee
- Because the merchant pays nothing
- Because the ECB taxes every payment

8. N26 (2023) earned more revenue per customer than Revolut (2024), and lost money. The main reason:

- its customers paid fewer fees
- interest rates were negative in 2023
- much higher costs per customer, for lack of scale
- the pound is worth more than the euro

SECTION 3.6 · BLOCK 3

🎤 TALK · 20 MIN

Crowdfunding, peer-to-peer lending, buy now pay later

Instructor's talk, or self-study reading.

FOUR FORMS OF CROWDFUNDING

What the person who puts in money gets back

- **Donation**: nothing in return (a school to rebuild, a cause).
- **Reward**: a product or a perk, often the first batch — it is a *pre-sale* (Kickstarter, in Switzerland wemakeit). An "all or nothing" campaign: if the target is not reached, nobody is charged.
- **Lending** (crowdlending): individuals lend to an SME or to another individual, with interest.
- **Equity** (equity crowdfunding): backers become shareholders; they hope for a capital gain on resale, in five to ten years, or nothing if the company fails.

The framework. In the EU, Regulation 2020/1503 (**ECSP**) gives lending and equity platforms an EU-wide licence, for offers of up to **€5 million** per project over twelve months, with a standard key information sheet and a knowledge test for non-sophisticated investors. In Switzerland, professionally accepting funds from more than 20 people normally requires a banking licence; platforms use the exceptions in the Banking Ordinance (funds on a **settlement account** passed on within 60 days) or the fintech licence (Day 5).

THE MECHANISM · A PEER-TO-PEER LOAN

Six steps, from application to repayment

▸ AN SME BORROWS CHF 300,000 FROM 600 INDIVIDUALS

1 · The application. The SME uploads its accounts and its project to the platform: CHF 300,000 over three years for a machine.

2 · The analysis. The platform (not the lenders) analyses the file, assigns a **risk grade** (A to E) and a rate, for example 6.5% for a grade B.

3 · The raise. The project is published; 600 individuals lend CHF 100 to 5,000 each. If the amount is not reached in time, the money is returned.

4 · The payout. The funds pass through an account of the platform or a partner bank — in Switzerland, a *settlement account* that must be emptied within 60 days — then are paid to the SME, minus the platform's commission (2 to 4% of the amount, paid by the borrower).

5 · The repayments. Each month the SME repays; the platform distributes to the 600 lenders, minus its servicing fees (often ~1% a year, paid by the lenders).

6 · Default. If the SME stops paying, the platform chases and collects on the lenders' behalf. The lenders bear the loss; the platform has already collected its commission — hence the incentive problem: it earns on volume, not on quality.

✓ All 6 steps are shown

Expected return ≈ (1 – PD) × rate – PD × LGD – platform fees

The advertised rate is not the return. A loan at 18% with an 18% PD and a 70% LGD yields, after 1% fees, about 1.2% — less than a loan at 9%. And one default in a portfolio of two loans is a disaster; in two hundred, a statistic.

SPLIT PAYMENT · BUY NOW, PAY LATER

"Pay in 4, interest-free": who pays, and is it credit?

- **The mechanism.** At checkout, the provider (Klarna, Afterpay, PayPal Pay in 4, in Switzerland also Powerpay or invoices payable in 30 days) runs a check in a few seconds, pays the merchant at once, and is repaid by the customer in several instalments, often every two weeks.
- **Who pays.** The **merchant**: a fee of a few percent, higher than a card, which it accepts because split payment raises basket size and conversion. The customer pays nothing if they pay on time — and **late fees** otherwise.
- **The status.** In the EU, the old directive excluded interest-free credit repayable within three months; the new one (2023/2225) brings it in from **20 November 2026**, with a creditworthiness assessment. In the UK, the FCA has regulated it since **15 July 2026**. In Switzerland, the KKG/LCC excludes credit repayable within **three months at most**; the

old “four instalments within twelve months” exception disappeared with the 2016 revision.

CHECK · 3 MIN

Which form of crowdfunding?

DONATION · REWARD · LENDING · EQUITY

400 people pay CHF 75 each to receive a bag from the first batch, made in Dakar

Donation Reward Lending Equity

An SME borrows CHF 300,000 at 4.5% over three years from 600 individuals

Donation Reward Lending Equity

120 investors together buy 8% of the shares of a young fintech

Donation Reward Lending Equity

A collection to rebuild a neighbourhood school after a flood

Donation Reward Lending Equity

Backers receive a T-shirt and their name on the project’s website

Donation Reward Lending Equity

One individual lends CHF 2,000 to another, through a platform, to buy a used car

Donation Reward Lending Equity

0 of 6 sorted

SECTION 3.7 · BLOCK 3

🛠 WORKSHOP · 55 MIN

A campaign, a loan portfolio, a debate

A · 10 MIN · THE CAMPAIGN

What is left of a successful campaign

A bag designer in Dakar runs a reward campaign: **400 backers** give **CHF 75** on average. The platform takes **5%**, payment processing **3%**; making the promised bags costs **30%** of the amount raised; shipping costs **CHF 5** per backer.

How much does the designer keep, in CHF, once the platform, payment processing, making the promised bags and shipping are paid?

… CHF Check

B · 20 MIN · THE PEER-TO-PEER LENDING SIMULATOR

Lend CHF 10,000: choose the risk grade, then the number of loans

One-year loans, repaid in one go. A platform grades borrowers from A (safe) to E (risky). If a borrower defaults, you recover 30% (LGD 70%). Platform fees: 1% a year. Each click runs **2,000 simulated years**; the chart shows the distribution of your return.

Risk grade

C · rate 9 % · PD 6 %

Number of loans (CHF 10,000 split equally)

10

GRADE	ADVERTISED RATE	PD	EXPECTED RETURN
A	4 %	1.5 %	to calculate ↓
B	6 %	3 %	to calculate ↓
C	9 %	6 %	to calculate ↓
D	13 %	11 %	to calculate ↓
E	18 %	18 %	to calculate ↓

Simulate 2,000 years →

Expected return on a **grade E** loan (rate 18%, PD 18%, LGD 70%, fees 1%), in %?

… % Check

Same calculation for **grade C** (rate 9%, PD 6%)?

… % Check

✍️ **Your best combination (grade, number of loans) and why. What does diversification change — to the average return, and to the risk of losing?**

The simulation log appears above; summarise it.

C · 25 MIN · STRUCTURED DEBATE

“Is buy now, pay later credit?”

The rules. Two sides drawn by lot: *“yes, and it must be regulated as such”* against *“no, it is a means of payment”*. Five minutes of preparation with the arguments below, two minutes of opening per side, six minutes of exchange, one minute of conclusion per side. Every argument must cite a mechanism or a regulator’s position.

▸ FIRST, SORT THE ARGUMENTS: WHICH SIDE DO THEY SERVE?

The provider pays the merchant at once; the customer repays later

“It is credit” “It is not credit”

No interest is charged to a customer who pays on time

“It is credit” “It is not credit”

A CHF 5 late fee on a CHF 25 instalment, two weeks late

"It is credit" "It is not credit"

It is the merchant, not the customer, who pays the fee

"It is credit" "It is not credit"

A customer can open plans with several providers without any of them seeing the others

"It is credit" "It is not credit"

The amounts are small and the term short: six weeks

"It is credit" "It is not credit"

EU Directive 2023/2225 brings deferred payment into consumer credit from 20 November 2026

"It is credit" "It is not credit"

In the UK, since July 2026 the FCA has required an affordability check before every plan

"It is credit" "It is not credit"

0 of 8 sorted

A CHF 5 late fee on a CHF 25 instalment paid 14 days late: what **annual cost** (simple), in %?

… % Check

Preparation 4:56 Pause ↺

One side's opening 1:56 Pause ↺

⏱ Exchange **5:56** Pause ↺

✍️ **Your side, the other side’s strongest argument, and your own position after the debate.**

e.g. “Yes” side. Their best argument: the merchant pays… My position: …

DEBRIEF · 15 MIN

The jury (the instructor, or a team drawn by lot) names the side that made the best use of the regulators’ positions. Then a question for the project: could your fintech offer buy now, pay later, and under which licence?

SECTION 3.8 · BLOCK 4

🎤 TALK · 15 MIN

Digital credit in Africa: effects and abuses

Instructor's talk, or self-study reading.

THE FOUNDING CASE · M-SHWARI, KENYA, 2012

A savings account and a loan in the M-Pesa menu

- **The product.** Launched by Commercial Bank of Africa and Safaricom: a savings account and a 30-day loan, from 100 shillings, opened from the M-Pesa menu, with no branch and no paper. The price: a **facilitation fee of 7.5%** of the amount, not an interest rate.
- **The score.** No credit file for most customers: the credit limit is computed from the M-Pesa history (tenure, deposits, transfers, savings on M-Shwari). The first limit is small; it grows with on-time repayments.
- **The measured effect.** Bharadwaj, Jack and Suri (*Journal of Development Economics*, 2021) compare customers just above and just below the eligibility threshold: a third of those eligible borrow; households with access to the loan are **6.3 points** less likely to forgo essential spending after a shock (illness, crop loss). But neither savings nor productive assets rise: digital credit helps absorb a blow, it does not lift people out of poverty.

IN WEST AFRICA

Credit and savings through the mobile wallet

- **Baobab** (a microfinance group, formerly MicroCred) launched **M-Baobab** in Senegal in 2022 with Orange Finances Mobiles: fully digital credit and savings, from the Orange Money app or the USSD code #144#. Baobab+ built a score from the payment history of solar kits paid for by mobile money.
- **The WAEMU framework.** An e-money institution cannot lend (Day 2): the credit is extended by a partner bank or microfinance institution, under the usury rate (14% for banks, 24% for microfinance institutions). That constraint is what sets the West African market apart from Kenya.

THE ABUSES

What went wrong, and what regulators did

- **The hidden price.** A “per loan” fee over 30 days looks modest; annualised, it often exceeds 90%.
- **Over-indebtedness and the blacklist.** In Kenya, borrowers were reported en masse to credit bureaus for defaults of a few hundred shillings, which then shut them out of bank credit.
- **Debt shaming.** Lending apps read the phone’s contact list and wrote to the late borrower’s contacts.
- **The response.** Kenya put digital lenders under central-bank supervision (Central Bank Act amended in 2021, **Digital Credit Providers Regulations** of March 2022): licence required, access to the contact list banned, threats and publication of data banned, no reporting to the credit bureau below 1,000 shillings. In 2025 the Kenyan press still reported harassment by licensed lenders.
- **The opaque score.** A refused customer does not know why, and so can neither challenge it nor improve.

CHECK · 4 MIN

Positive effect, no effect, or abuse?

▸ WHAT RESEARCH AND REGULATORS HAVE DOCUMENTED

Households with access to M-Shwari forgo essential spending less often after a shock

Positive effect | No effect | Abuse

Access to M-Shwari and households’ savings or productive assets

Positive effect | No effect | Abuse

Apps wrote to the phone contacts of a borrower who was late

Positive effect | No effect | Abuse

Borrowers reported to credit bureaus for a few hundred unpaid shillings

Positive effect | No effect | Abuse

Access to M-Pesa and poverty in Kenya (Day 2)

Positive effect | No effect | Abuse

A credit refusal given with no reason at all, by a model nobody can explain

Positive effect | No effect | Abuse

0 of 6 sorted

M-Shwari: a 7.5% fee for a 30-day loan. **Simple annual cost**, in %?

… % Check

And if the borrower takes a new loan as soon as the previous one is repaid, all year long: **compound annual cost**, in %?

… % Check

SECTION 3.9 · BLOCK 4

PROJECT AND PEER REVIEW · 75 MIN

Your app's pension-savings plan

35 MIN · IN TEAMS · THE PRODUCT SHEET

Design the plan, its fees, and what makes it trustworthy

Both teams' project: **an app that lets people in Senegal, Congo or a neighbouring country save for their retirement** by paying small amounts every month until 65, **invested in Swiss blue-chip equity funds**. The app and the company must be transparent and earn trust. Set the simulator with *your* assumptions, then fill in the sheet. Each member fills in their notebook — the same version for the whole team.

WHAT A SMALL MONTHLY PAYMENT BECOMES AT 65 — AND WHAT THE FEES TAKE

Payment per month

10000

Customer's currency

FCFA

Age at the first payment

30

Fund's annual return before fees (%) — an assumption

5

Fund fees, % a year

0.2

Your app's fee, % of assets a year

0.8

Transfer and exchange, % of each payment

1

Swiss franc against the customer's currency, % a year

0

4,200,000 FCFA
PAID IN FROM AGE 30 TO 65

8,846,806 FCFA
CAPITAL AT 65, AFTER ALL FEES

11,129,790 FCFA
THE SAME PLAN WITH NO FEES AT ALL

20.5 %
SHARE OF THE CAPITAL TAKEN BY FEES

6,192,764 FCFA
IF SHARES LOSE 30% IN THE YEAR YOU TURN 65

Payments at the start of each month, return compounded monthly, annual fees deducted from the return. The return is an assumption, not a promise: Swiss large caps also lost more than a third in 2008 and about a quarter within weeks in 2020. The Swiss franc gained value against the euro, and so against the FCFA,

over twenty years (about 1.60 francs to the euro in 2007, 0.93 in 2025), but nothing guarantees what comes next. These are the figures the app must show the customer.

1. The plan: for whom, how much a month and in which currency, through which channel they pay in (Wave, Orange Money, transfer…), into which fund exactly, all the fees (with the share of capital computed above), how the money is paid out at 65 — and what happens if the customer has to withdraw earlier.

e.g. For women traders aged 25 to 45 in Dakar; from 5,000 FCFA a month via Orange Money; an index fund of Swiss large caps (which one, with its fees); app fee …% a year, i.e. …% of the capital at 65; at 65, paid out as …; early withdrawal: …

2. Transparency and trust: who holds the money and the fund units (custodian bank, segregated account), what the customer sees in the app each month (balance, fees in francs, performance), how they can check it without taking your word for it, and the sentence the app shows them the day the fund has lost 20%.

e.g. The units are held in each customer's name at a Swiss custodian bank, separate from the company's accounts; monthly statement: paid in, fees in FCFA, value…; the customer can…; message after a fall: "…"

3. The safeguards: against misappropriation of the funds, against mis-selling (promising a return, pushing a customer who needs their money), against currency risk, against payments being abandoned. One measure for each.

e.g. No return promised, only scenarios…; an external auditor…; a missed payment costs nothing and loses nothing…; …

25 MIN · PEER REVIEW

Another team reads your sheet; you read theirs

Swap screens between the two teams. Read the other team's three fields and answer the five questions below *for them*; tell them your answers out loud. Then write down what your own team received.

› THE REVIEW GRID · THE TEAM YOU ARE REVIEWING

The customer is precise: who saves, how much a month, from what age, and why they would do it rather than the tontine.

Yes | To be clarified | No

Fees are given in francs (FCFA, CDF…), not only in %, and their share of the capital at 65 is calculated.

Yes | To be checked | No

We know who holds the money and the fund units, and the customer can check it themselves.

Yes | To be clarified | No

The customer is warned, with a figure, that one year can lose 30% — and knows what happens then.

Yes | Partly | No

At least one real safeguard against misappropriation of funds or mis-selling.

Yes | Too weak | No

0 of 5

Reading and grid **9:56** Pause ↺

4. The criticism received from the other team, and what you change in your product.

e.g. The other team: our fees take 28% of the capital at 65… → app fee cut to 0.5% a year.

BEFORE DAY 4 · FRIDAY 30 OCTOBER

- Day 4 is about blockchain, crypto-assets and decentralised finance. You will create a wallet on a **test** blockchain: nothing to buy, nothing to install in advance; an up-to-date Chrome, Firefox or Brave browser is enough.

- For the project: note whether any part of your app (proof that the units exist, the customer's statement, the transfer from Africa) *could* run on a blockchain. Day 4 will make you say whether it really should.

DAY 4 · FRIDAY 30 OCTOBER 2026 · MFB 545

Blockchain, crypto-assets, DeFi and tokenisation

A blockchain is a ledger that nobody keeps alone and everybody can verify. This day shows it for real: you build a blockchain among yourselves, then you read on DefiLlama how much money decentralised finance carries today, where, and what it loses. Then you borrow and swap on protocols with no bank, you split a building into tokens, you see how hundreds of millions get stolen — and you decide whether your team's pension-savings app needs a blockchain, or only a proof that the money is there.

🕘 **09:00–12:15 · 13:30–16:45 · four 90-minute blocks**

💻 **laptop · an up-to-date Chrome, Firefox or Brave**

🎤 **the talk is short — 🛠️ most of the time, you are the one working**

THE DAY'S WORKSHOP · THE CLASS BLOCKCHAIN

Everyone in the room is a node: make your keys, sign, hash, mine, pay someone, then cheat — and see why cheating fails. Section 4.1 B, from the first block.

⛓️ **Open the class blockchain ↗** · 🖥️ **The room screen ↗**

The brief · section 4.1 B

The afternoon: blockchains and DeFi in numbers, on DefiLlama — a slide deck (in French), then the section 4.2 workshop, which reads the same data live.

🧱 **The slides ↗** · **The workshop · section 4.2**

BY THE END OF DAY 4, YOU WILL BE ABLE TO

- Explain in three ideas why a shared ledger holds without a trusted third party — signature, fingerprint, consensus — and what it does not guarantee.
- Explain, having done it among yourselves, why a shared ledger holds without a keeper: keys, hashes, mining, consensus — and what it does not prove.

- Read DefiLlama: TVL, the chains that matter, the protocol families, the weight of stablecoins, who earns the fees and how billions get lost.
- Calculate the health factor of a DeFi loan, its liquidation price, and the price obtained on an automated exchange.
- Tell apart the three families of stablecoins and say what backs each one.
- Place tokenisation and central bank digital currencies in Switzerland (DLT Act, SDX, Helvetia) and in WAEMU.
- Decide, with a published method, whether your project needs a blockchain — and prove to a customer that their units exist.

BEFORE YOU START — TICK

- ☐ **Signed in** on this site (the button at the bottom right).
- ☐ **An up-to-date browser** (Chrome, Firefox or Brave): the class blockchain and DefiLlama run in it, nothing to install.
- ☐ **A calculator** (the phone's is enough): shares, health factors and swap prices are worked out by hand before the page checks them.
- ☐ **Your team's pension-savings plan** (Day 3, section 3.9): the day ends on it.

BLOCK 1 · 09H00–10H30 why "trustless" works · build your blockchain · the DefiLlama tour

4.1 **Why "trustless" works**
Signature, fingerprint, consensus · the life of a transaction · what the chain guarantees
🎤 TALK · 20 MIN

4.1 B **Build your blockchain**
Every student is a node · keys, hashes, mining, payments, forks, cheating · what the ledger proves
🛠 WORKSHOP · 60 MIN

4.2 **The DefiLlama tour: decentralised finance in numbers**
Live TVL · chains and protocols · stablecoins · fees, revenue, hacks

🛠 WORKSHOP · 55 MIN

BLOCK 2 · 10:45–12:15 lend, swap, stabilise · the DeFi labs · the quiz

4.3 **Lend, swap, stabilise — with no bank**

Health factor and liquidation · x × y = k · three families of stablecoins

🎤 TALK · 15 MIN

4.4 **DeFi labs · borrow, swap**

The loan and its liquidation price · price impact · the DeFi capstone

🛠 WORKSHOP · 50 MIN

4.5 **End-of-morning quiz**

Eight questions · launched in class · counts towards participation

✅ QUIZ · 10 MIN

BLOCK 3 · 13:30–15:00 tokenisation, Switzerland and WAEMU · “do you need a blockchain?”

4.6 **Tokenisation and central bank digital currencies**

A token, a right · DLT Act, SDX, Helvetia · the BCEAO and the eNaira

🎤 TALK · 20 MIN

4.7 **Split a building · do you need a blockchain?**

The fractionalisation simulator · the decision tree applied to your app

🛠 WORKSHOP · 55 MIN

BLOCK 4 · 15:15–16:45 MEV, hacks, MiCA · prove the money is there · DeFi Millionaire

4.8 **MEV, three hacks, MiCA**

The sandwich attack in numbers · The DAO, Ronin, Bybit · MiCA, FINMA, BCEAO

🎤 TALK · 15 MIN

4.9 **Prove the money is there · DeFi Millionaire**

The customers' Merkle tree · your app's proof and keys · the game

🛠 WORKSHOP · 75 MIN

← Course page

reset my Day 4 progress on this device

SECTION 4.1 · BLOCK 1

🎤 TALK · 20 MIN

Why “trustless” works

The instructor’s talk. If it does not take place, read this section on your own, then do the sorting at the bottom of the page.

THE PROBLEM

A ledger that nobody keeps alone

Your bank account is a line in the bank’s ledger. You believe the balance because you trust the bank, its auditor and FINMA. A blockchain replaces that third party with a ledger **copied on thousands of computers**, which anyone can reread from day one. Three questions must then be answered with no one in charge: who may spend? can the past be rewritten? in what order do transactions go through? Three ideas, one per question.

- **1 · The signature — who can spend.** Each account is a pair of keys. The **private** key signs; the **public** key (from which the address comes) lets everyone check the signature without ever knowing the private key. Without a valid signature, no computer on the network accepts the transaction. That is the sheet public key and private key.
- **2 · The fingerprint — the past is not rewritten.** Each block contains the **fingerprint** (the hash) of the previous block. Changing one comma in an old block changes its fingerprint, and therefore that of every block after it: the forgery shows immediately. That is the sheet what is a hash?.
- **3 · Consensus — who writes the next block.** Bitcoin draws lots by computation (*proof of work*, one block every ten minutes or so). Ethereum, since September 2022, by stake (*proof of stake*): validators each lock up 32 ETH; every 12 seconds one of them proposes a block and the others attest it. A validator who cheats loses part of its stake. After two “epochs” (about 13 minutes) the block is **finalised**: undoing it would cost at least a third of all the stake.

These three ideas are enough. The rest — smart contracts, DeFi, tokens — is code that runs on this ledger and inherits its guarantees, *and its limits*.

THE MECHANISM · THE LIFE OF A TRANSACTION

Seven steps between “Send” and the public ledger

▸ 0.001 ETH SENT TO A CLASSMATE ON ETHEREUM

1 · The wallet builds the transaction. Recipient, amount, a counter (the *nonce*: this is your 1st, 2nd… transaction, which prevents it being replayed), and two caps: the maximum **gas** (the computing work allowed; a simple transfer uses exactly 21,000) and the maximum price per unit of gas.

2 · It signs it with the private key. The key never leaves your device; only the signature goes out. Anyone can check, with your public key, that it really is you and that nothing has been altered.

3 · It goes into the public queue. The *mempool*: thousands of nodes receive it and check the signature and the balance. It is visible to everyone *before* it is executed — remember this for section 4.8.

4 · A validator picks it. Every 12 seconds, a validator drawn by lot proposes a block; it orders the transactions, usually those offering the best tip first.

5 · It is executed and paid for. The balance changes. Fee = gas used × gas price. Since 2021 (EIP-1559), the "base" part of that price is **destroyed**; only the tip goes to the validator.

6 · It becomes final. The other validators attest the block; after two epochs, about 13 minutes, it is **finalised**. No customer service can cancel it or refund it.

7 · Anyone can read it. A block explorer (Etherscan) is just a website that reads the public ledger: sender, recipient, amount, fee, block, time. Your name is not there — but your address, once linked to you, reveals your whole history.

✓ All 7 steps are shown

CHECK · 5 MIN

What the blockchain guarantees — and what it does not

› GUARANTEED BY THE CHAIN · NOT GUARANTEED: SOMEONE ELSE IS NEEDED

That the transaction was authorised by the holder of the private key

Guaranteed by the chain | Not guaranteed

That a finalised transaction will not be altered

Guaranteed by the chain | Not guaranteed

That the building represented by a token exists and is not mortgaged twice

Guaranteed by the chain | Not guaranteed

That a smart contract does what is written in its code

Guaranteed by the chain | Not guaranteed

That an address really belongs to the person who claims it

Guaranteed by the chain | Not guaranteed

That funds can be recovered after losing the secret phrase

Guaranteed by the chain | Not guaranteed

That the balance shown by a stablecoin is covered by dollars at the bank

Guaranteed by the chain | Not guaranteed

0 of 7 sorted

SECTION 4.1 B · BLOCK 1

🛠 WORKSHOP · 60 MIN

Build your blockchain

A · 45 MIN · THE CLASS BLOCKCHAIN

Everyone in the room is a node

Section 4.1 says why a shared ledger holds without a trusted third party. Before looking at the real ones (section 4.2), build one among yourselves: your browser keeps its own copy, checks every block and every payment, and refuses what is false. The site only carries the messages. Seven rounds: keys and signature, hash and block, mining, payments, the chain, cheating, the project.

⛓ Open the class blockchain ↗ 🖥 The room screen ↗

▸ WHAT THE WORKSHOP RECORDED ON THIS DEVICE

- ☐ Key pair made (round 1)
- ☐ Message signed and verified (round 1)
- ☐ A nonce found by hand (round 2)
- ☐ A block mined on the class chain (round 3)
- ☐ A payment signed and sent (round 4)
- ☐ My copy altered — and refused by the others (round 6 A)
- ☐ A double spend attempted (round 6 B)
- ☐ My copy repaired

The workshop page opens in its own tab; this list updates when you come back here.

B · 15 MIN · FOR THE PROJECT

Which entry of your app deserves this ledger?

What you just built guarantees three things: a signed entry cannot be forged on the customer’s behalf; a past entry cannot be rewritten without everyone seeing it; no single

keeper can alter the ledger. It does not guarantee that the units exist at the custodian — the chain proves the entry, not the real world (section 4.9).

✍️ **One entry of your app that would gain from a shared ledger (which one, and who the nodes would be); one that does not need it (and why an ordinary database is enough); and what the ledger will never prove.**

e.g. Each customer's unit statement: nodes = our company, the Swiss custodian, an auditor, the regulator…

DEBRIEF · 10 MIN

On the room screen: the nodes, the longest chain, the balances. Who won the most blocks, and why? What happened to the payment spent twice? And when the instructor altered the site's copy, what did your nodes do?

SECTION 4.2 · BLOC 1

🛠️ ATELIER · 55 MIN

The DefiLlama tour: decentralised finance in numbers

THE WORKSHOP RULE

DefiLlama ↗ is the public dashboard of decentralised finance: free, no account, no ads, open source. It reads the contracts of more than 400 blockchains and adds up what they hold. This page reads the same data (live) and asks you questions about it; open DefiLlama alongside and check every figure at the source.

Two words first. **TVL** (*total value locked*) is the value, in dollars, of the tokens deposited in the contracts of a protocol or a chain: the money at work, not the money changing hands. A **protocol** is a program deployed on a chain (Aave, Uniswap, Lido); a **chain** is the ledger that carries it (Ethereum, Solana, Tron).

Open DefiLlama ↗

🧱 The slides: blockchains and DeFi in numbers (French) ↗️

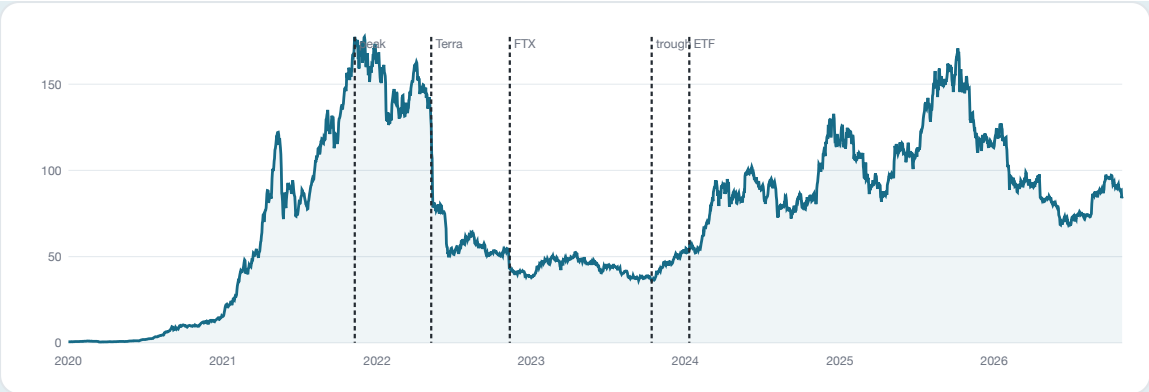
A · 10 MIN · THE MAP

How much money, where, and since when

85.1 Md $	420	5,489	51.8 %
TVL OF ALL DEFI	CHAINS TRACKED	DEFI PROTOCOLS (30 OCT)	ON ETHEREUM

77,451 $	2,249 $
ONE BITCOIN	ONE ETHER

▸ THE TVL OF ALL DEFI SINCE 2020, IN BILLIONS OF DOLLARS



Move the mouse over the curve. The marks: the peak of November 2021, the Terra collapse (May 2022), the FTX bankruptcy (November 2022), the trough of October 2023, the US bitcoin ETFs (January 2024).

What share of total TVL is on **Ethereum**, in % (read the two figures above)?

… % Check

How far did TVL fall between the **peak** (2021-11-09, 177.5 bn) and the **trough** that followed (2023-10-13, 36.0 bn), in %?

… % Check

B · 15 MIN · CHAINS AND PROTOCOLS

Who carries the money, and who puts it to work

▸ THE FIFTEEN LARGEST CHAINS BY TVL

1 · Ethereum	44.1 Md · 51.8 %
2 · Bitcoin	6.1 Md · 7.2 %
3 · Base	5.9 Md · 6.9 %
4 · Solana	5.5 Md · 6.4 %
5 · Tron	5.4 Md · 6.4 %
6 · BSC	5.2 Md · 6.2 %
7 · Hyperliquid L1	2.6 Md · 3.1 %
8 · Arbitrum	1.4 Md · 1.7 %
9 · Robinhood Chain	1.1 Md · 1.3 %
10 · Monad	0.9 Md · 1.0 %
11 · Polygon	0.7 Md · 0.8 %

12 · Arc		0.6 Md · 0.7 %
13 · Avalanche		0.5 Md · 0.6 %
14 · Plasma		0.5 Md · 0.6 %
15 · Sui		0.5 Md · 0.5 %

The first 1 chain(s) carry 51.8 % of TVL · slide

› THE TWENTY LARGEST DEFI PROTOCOLS, CENTRALISED EXCHANGES EXCLUDED · SNAPSHOT OF 30 OCTOBER 2026

#	PROTOCOL	CATEGORY	CHAINS	TVL
1	**Lido**	Liquid staking	5	21.7 Md $
2	**LayerZero V2**	Bridge	91	15.7 Md $
3	**Aave V3**	Lending	21	14.6 Md $
4	**SSV Network**	Staking pool	1	11.7 Md $
5	**Morpho Blue**	Lending	45	10.5 Md $
6	**WBTC**	Bridge	1	9.0 Md $
7	**Binance staked ETH**	Liquid staking	2	8.6 Md $
8	**Coinbase Bridge**	Bridge	5	7.7 Md $
9	**Hyperliquid Bridge**	Bridge	2	6.6 Md $
10	**EigenCloud**	Restaking	1	5.8 Md $
11	**Binance Bitcoin**	Bridge	1	5.3 Md $
12	**Sky Lending**	Collateralised stablecoin (CDP)	1	5.1 Md $
13	**SparkLend**	Lending	2	5.1 Md $
14	**Ethena USDe**	Basis trading	1	4.5 Md $
15	**ether.fi Stake**	Liquid staking	6	4.3 Md $
16	**JustLend V1**	Lending	1	3.6 Md $
17	**Arbitrum Bridge**	Canonical bridge	1	3.3 Md $
18	**JustLend sTRX**	Liquid staking	1	3.3 Md $

19	**Babylon Protocol**	Restaking	1	3.2 Md $
20	**USDT0**	Bridge	1	3.1 Md $

All DeFi protocols together: 262.8 bn $ (some is counted twice when one protocol deposits into another). 4,276 protocols weigh under one million; 51 exceed one billion. DefiLlama also ranks the reserves of centralised exchanges (Binance, OKX…); they are left out here: a centralised exchange is not decentralised finance.

How many chains, starting from the first, carry **90%** of TVL? (the slider above)

… chains Check

The **top ten protocols** in the table weigh what share of all protocols' TVL (262.8 bn $), in %?

… % Check

› WHICH FAMILY DOES EACH PROTOCOL BELONG TO?

Aave V3

Lending | Liquid staking | Exchange (AMM) | Bridge | Collateralised stablecoin

Lido

Lending | Liquid staking | Exchange (AMM) | Bridge | Collateralised stablecoin

Uniswap

Lending | Liquid staking | Exchange (AMM) | Bridge | Collateralised stablecoin

LayerZero

Lending | Liquid staking | Exchange (AMM) | Bridge

Collateralised stablecoin

Sky (ex-MakerDAO)

Lending | Liquid staking | Exchange (AMM) | Bridge | Collateralised stablecoin

Morpho

Lending | Liquid staking | Exchange (AMM) | Bridge | Collateralised stablecoin

0 of 6 sorted

C · 10 MIN · STABLECOINS

Three hundred billion dollars that are not dollars

299.6 Md $	334	60.0 %	91.1 %
ALL STABLECOINS	STABLECOINS TRACKED	USDT (TETHER)	BACKED BY RESERVES

3.5 ×
DEFI'S TVL

▸ THE TEN LARGEST STABLECOINS, WITH THEIR CURRENT PRICE

#	TOKEN	BACKING	CIRCULATING	SHARE	PRICE
1	**Tether** · USDT	fiat-backed	179.66 Md $	60.0 %	0.9993 $
2	**USD Coin** · USDC	fiat-backed	71.84 Md $	24.0 %	0.9999 $
3	**Sky Dollar** · USDS	crypto-backed	6.45 Md $	2.2 %	1.0005 $
4	**Dai** · DAI	crypto-backed	4.73 Md $	1.6 %	1.0002 $

5	**Ethena USDe** · USDe	crypto-backed	4.50 Md $	1.5 %	0.9986 $
6	**World Liberty Financial USD** · USD1	fiat-backed	3.89 Md $	1.3 %	0.9982 $
7	**Global Dollar** · USDG	fiat-backed	2.92 Md $	1.0 %	1.0002 $
8	**PayPal USD** · PYUSD	fiat-backed	2.82 Md $	0.9 %	1.0003 $
9	**Ripple USD** · RLUSD	fiat-backed	2.42 Md $	0.8 %	1.0001 $
10	**BlackRock USD** · BUIDL	fiat-backed	2.01 Md $	0.7 %	1.0000 $

None of the ten is more than half a cent from the dollar right now. March 2023: USDC fell to $0.88 over a weekend, because 3.3 billion of its reserves sat at Silicon Valley Bank. May 2022: Terra's UST fell to zero.

USDT and USDC together: what share of all stablecoins, in %?

… % Check

D · 10 MIN · WHAT IT EARNS, WHAT IT LOSES

Fees, revenue and hacks

› FEES PAID BY USERS AND REVENUE KEPT BY THE PROTOCOL, OVER 30 DAYS · SNAPSHOT OF 30 OCTOBER 2026

PROTOCOL	CATEGORY	FEES 30 D	REVENUE 30 D	KEPT
Tether	Stablecoin issuer	522.4 M $	522.4 M $	100 %
Circle USDC	Stablecoin issuer	209.0 M $	209.0 M $	100 %
PumpSwap	Exchange (AMM)	105.9 M $	14.0 M $	13 %
Polymarket US	Prediction market	86.5 M $	0.0 M $	0 %

Hyperliquid Perps	Derivatives	80.9 M $	64.1 M $	79 %
Papertrade	Derivatives	67.9 M $	67.9 M $	100 %
pump.fun	Launchpad	61.9 M $	44.8 M $	72 %
Uniswap V4	Exchange (AMM)	54.9 M $	0.0 M $	0 %
Lido	Liquid staking	49.4 M $	3.0 M $	6 %
Aave V3	Lending	36.2 M $	4.6 M $	13 %
Uniswap V3	Exchange (AMM)	33.0 M $	7.9 M $	24 %
Polymarket International	Prediction market	32.4 M $	15.9 M $	49 %
Hyper Foundation HYPE Staking	Staking pool	31.9 M $	1.0 M $	3 %
Flap sh	Launchpad	29.5 M $	7.0 M $	24 %

“Fees”: what users paid. “Revenue”: the share the protocol keeps for its treasury or its token holders; the rest goes to depositors, validators or market makers.

WHERE DO THE FEES GO?

Tether: 522 M fees, 522 M revenue. The fees are the interest on the Treasury bills backing USDT.

To depositors / market makers | To the protocol and its token holders | Shared

Uniswap V4: 55 M fees, 0 revenue.

To depositors / market makers | To the protocol and its token holders | Shared

Hyperliquid (derivatives): 81 M fees, 64 M revenue.

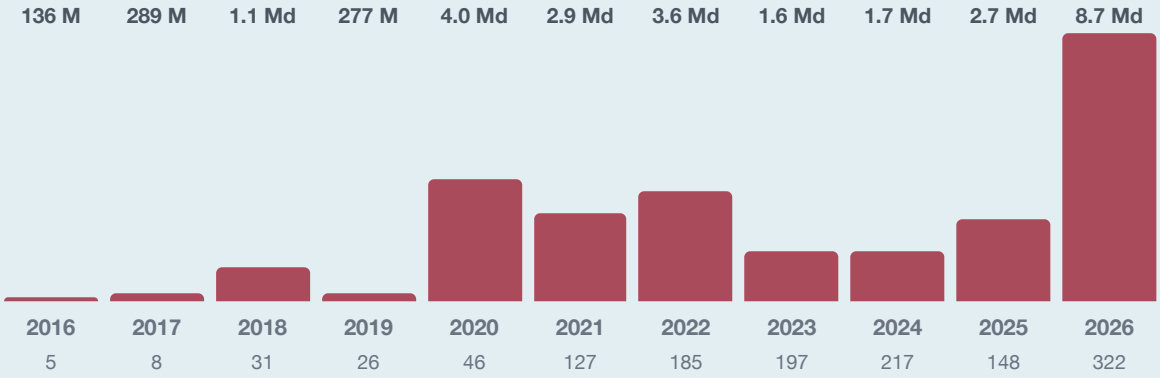
To depositors / market makers | To the protocol and its token holders | Shared

Lido: 49 M fees, 3 M revenue.

To depositors / market makers | To the protocol and its token holders | Shared

0 of 4 sorted

HACKS SINCE 2016, IN MILLIONS OF DOLLARS A YEAR



TARGET	DATE	AMOUNT	HOW
Binance CEX	2026-10-29	6.36 Md $	Frontend & infrastructure · Key Leaked via Infrastructure
LuBian	2020-12-28	3.50 Md $	Key compromise · Weak Key Generation
Bybit	2025-02-21	1.40 Md $	Social engineering · Signer Phishing
Ronin Bridge	2022-03-23	624 M $	Key compromise · Validator Key Compromised
Poly Network	2021-08-10	611 M $	Access control · Improper Access Control
Binance Bridge	2022-10-06	570 M $	Bridges & cross-chain · Proof Verifier Bug
Coincheck	2018-01-26	534 M $	Key compromise · Hot Wallet Key Compromised
Mt. Gox	2014-02-07	470 M $	Key compromise · Private Key Compromised

By type, all time: Key compromise 33 % · Frontend & infrastructure 27 % · Social engineering 11 % · Access control 7 % · Bridges & cross-chain 6 % · Token & share accounting 5 % · Oracle Manipulation 3 %

1,330 incidents listed by DefiLlama, 27.6 bn $ in all. Under each bar: the number of incidents that year. On 29 October 2026 Binance lost 6.3 billion dollars in twenty

minutes: a compromised laptop, withdrawals signed with the real keys. The ledger worked exactly as designed.

What share of all money ever stolen went through **key compromise** (the “By type” line), in %?

… % Check

E · 5 MIN · FOR THE PROJECT

Three figures for your dossier

✍️ **Three figures read today on DefiLlama that you could cite in the dossier of your pension-savings app — with, for each, what it proves or what it warns of (the size of stablecoins, the share of hacks due to keys, the yield on a USDC deposit…).**

1. … (source: DefiLlama, 30 October 2026) — what it proves: …

✍️ **If your clients’ fund units were tokens: on which chain, and why that one? Three criteria drawn from what you have just seen: what it already carries, what it costs, what has happened to it.**

Chain: … Because: 1. … 2. … 3. …

DEBRIEF · 5 MIN

Three questions: what does TVL measure, and what does it not? Why do stablecoins weigh more than all of DeFi together? The largest theft in the history of crypto-assets, yesterday, hit a centralised exchange, not a contract: what does that change about the “trustless” promise?

SECTION 4.3 · BLOCK 2

🎤 TALK · 15 MIN

Lend, swap, stabilise — with no bank

The instructor's talk, or read on your own.

LENDING · AAVE-TYPE

A loan with no file: the collateral does everything

A lending protocol is a program (a *smart contract*) that holds a common pool. Nobody runs it day to day: lenders deposit tokens into it and earn interest; borrowers deposit collateral into it and borrow against that. The program knows no name, no income, no history: no score, no collection, only collateral **worth more than the loan** and a liquidation rule that applies by itself.

Below, two people who do not know each other, a lender and a borrower, use the same contract. At each step: who calls the contract, what the contract does, and the state of the pool. The figures are those of the tool in section 4.4 (LTV 80%, threshold 82.5%, bonus 5%).

▸ TWO USERS, ONE CONTRACT · STEP BY STEP

1 · Awa deposits. Awa has 10,000 USDC sitting idle. She calls *supply* on the contract: the 10,000 USDC move from her address to the pool, and the contract writes her 10,000 aUSDC in return, a receipt whose balance grows with the interest. She does not know who will borrow.

```
Pool: 10,000 USDC deposited · 0 lent · utilisation 0% · rate 0%
```

2 · Moussa deposits his collateral. Moussa has 2 ETH (3,000 USDC each) and needs 3,000 USDC for six months, without selling his ETH. He calls *supply* with 2 ETH: the contract locks them as collateral, worth 6,000 USDC according to the price oracle. He can borrow up to 80%: 4,800 USDC.

```
Moussa's collateral: 2 ETH = 6,000 USDC · maximum borrow 4,800 USDC
```

3 · Moussa borrows. He calls *borrow* for 3,000 USDC. The contract checks the rule (3,000 ≤ 4,800), records a debt of 3,000 USDC against his address and sends him 3,000 USDC: Awa's. Nobody read a file; Awa signed nothing.

```
Pool: 10,000 deposited · 3,000 lent · utilisation 30% · borrow rate 4% a year · deposit rate 1.2%
```

```
Health factor: 6,000 × 0.825 ÷ 3,000 = 1.65
```

4 · For six months. Moussa uses his USDC: he swaps them for francs, pays a supplier, or buys more ETH. Every second, the contract increases his debt at the current rate, and Awa's aUSDC balance by the share that is hers. If others borrow, utilisation rises and both rates with it.

```
After six months at 4%: debt = 3,000 × (1 + 0.04 × ½) = 3,060 USDC · Awa's aUSDC ≈ 10,060
```

5a · Repayment, if all goes well. Moussa calls *repay* with 3,060 USDC: his debt drops to zero, and *withdraw* returns his 2 ETH, which he never sold. Awa calls *withdraw*: her 10,060 aUSDC are burned and 10,060 USDC return to her address. Neither of them spoke to anything but the program.

```
Pool: 0 lent · Awa + 60 USDC · Moussa − 60 USDC, 2 ETH recovered
```

5b · The fall, otherwise. In month four, ETH drops to 1,750 USDC. Moussa's collateral is now worth only 3,500 USDC against a debt of 3,040: health factor 3,500 × 0.825 ÷ 3,040 = **0.95**. The contract does nothing by itself, it has no "sell" button: it simply makes the position liquidatable by anyone.

```
Liquidation price: 3,040 ÷ (2 × 0.825) = 1,842 USDC per ETH · ETH is below it
```

6 · A third user liquidates. A bot, anybody's, watches the health factors. It calls *liquidationCall*: it repays half of Moussa's debt, 1,520 USDC, and the contract gives it 1,520 × 1.05 = 1,596 USDC worth of ETH, i.e. 0.912 ETH, taken from the collateral. The bot earns 76 USDC; Awa had nothing to do.

```
Moussa: 1.088 ETH left · debt 1,520 USDC · factor 1.088 × 1,750 × 0.825 ÷ 1,520 = 1.03
```

7 · The outcome. Moussa keeps 1.088 ETH and still owes 1,520 USDC: the 76 USDC bonus went in a few seconds, with no notice and no call. Awa was protected without doing anything: that is what over-collateralisation and public liquidation are for, not judging Moussa. What can still go wrong: a crash faster than the bots (the collateral falls below the debt: bad debt), an oracle giving a wrong price, a bug in the contract. When many loans are liquidated at once, the bots resell the seized ETH, which pushes the price down further: the **liquidation cascade**.

✓ All 8 steps are shown

The four rules the contract applies.

- **Maximum loan-to-value (LTV).** At opening, you borrow at most 80% of the collateral's value.
- **Liquidation threshold.** If the debt exceeds 82.5% of the collateral's value, the loan becomes liquidatable.
- **Health factor.** Collateral value × threshold ÷ debt. Above 1, nothing happens; below 1, anyone can repay part of the debt in the borrower's place and take some of their collateral with a 5% **bonus**. Automatic, public, no judge.
- **The rate.** It follows the pool's **utilisation** (the share lent out): the more of the pool is lent, the higher the rate, to attract deposits and slow borrowing. The interest borrowers pay goes to the depositors, less a reserve the protocol keeps.

Health factor = collateral × liquidation threshold ÷ debt

What it is used for, in practice.

- **Liquidity without selling.** Moussa needs 3,000 USDC for six months but wants to keep his ETH: he posts it as collateral instead of selling it, and gets it back as it was. No sale, no realised gain, no file, about twenty seconds.
- **A return on idle cash.** Awa places stablecoins she is not using; the rate is public, it moves every block, and she withdraws when she wants, as long as the pool is not fully lent out.
- **Leverage.** Borrow USDC against ETH to buy more ETH, and deposit that too: the position rises faster, and gets liquidated faster.
- **What it is not.** Credit for someone who has nothing: you must already hold more than you borrow. The rate is not fixed, the contract can have a bug, and a crash faster than the liquidators leaves **bad debt** that the protocol has to absorb.

Aave's real parameters vary from one asset to another and from month to month; the ones on this page are orders of magnitude, chosen for the calculation.

THE EXCHANGE · UNISWAP-TYPE

No order book: a formula

An **automated market maker** (AMM) is a pot holding two tokens, for example 100 ETH and 300,000 USDC. The program accepts any swap that leaves the product of the two quantities unchanged: **x × y = k**. The price is the ratio of the two reserves (300,000 ÷ 100 = 3,000 USDC per ETH). The more you buy at once, the more you empty the pot on one side and the higher the average price: that is **price impact**. Each swap pays a fee (often 0.3%) to the pot's depositors. If the pot's price drifts from the market's, arbitrageurs bring it back by swapping — and earn the gap.

Received = y × Δx × (1 − fee) ÷ (x + Δx × (1 − fee))

STABLECOINS

A token worth one dollar — provided that…

- **Backed by reserves.** A company issues one token for each dollar received and places the dollars in cash and Treasury bills (Circle's USDC, Tether's USDT). The value holds as long as the reserves exist and can be reached. In March 2023, 3.3 billion dollars of USDC's reserves were stuck at Silicon Valley Bank: USDC fell below 0.90 dollar for a weekend.
- **Backed by crypto-assets.** DAI is created by borrowers who deposit more collateral than they issue, with automatic liquidation — Aave's mechanism, applied to a currency.
- **Algorithmic.** No reserves: the peg holds through a swap mechanism with a second token. TerraUSD (UST) worked this way; in May 2022 a wave of selling sent both tokens to almost zero, and tens of billions of dollars vanished within days.

▸ WHAT BACKS THE VALUE OF THIS TOKEN?

USDC (Circle)

Currency reserves | Crypto-asset collateral | Algorithm

USDT (Tether)

Currency reserves | Crypto-asset collateral | Algorithm

DAI

Currency reserves | Crypto-asset collateral | Algorithm

TerraUSD (UST), until May 2022

Currency reserves | Crypto-asset collateral | Algorithm

EURC (Circle), a euro token

Currency reserves | Crypto-asset collateral | Algorithm

0 of 5 sorted

SECTION 4.4 · BLOCK 2

WORKSHOP · 50 MIN

DeFi labs · borrow, swap

A · 15 MIN · THE LOAN

2 ETH as collateral, USDC borrowed, and the ETH price falling

First write your prediction: at what ETH price will this loan be liquidated? Then move the price. Maximum LTV 80%, liquidation threshold 82.5%, liquidation bonus 5%.

ETH deposited as collateral

2

USDC borrowed

3000

ETH price, in USDC

6,000	4,800	1.65	1,818.18
COLLATERAL VALUE, USDC	MAXIMUM LOAN AT 80%	HEALTH FACTOR	ETH LIQUIDATION PRICE

Healthy. The ETH price can still fall by 39.4% before liquidation.

2 ETH at 3,000 USDC as collateral, 3,000 USDC borrowed, liquidation threshold 82.5%. The **health factor**, to two decimals?

… Check

At what **ETH price** does this loan become liquidatable? Rounded to the nearest unit.

… USDC Check

At 1,700, a bot repays 1,500 USDC of the debt, with a 5% bonus. How much **bonus does the borrower lose**, in USDC?

… USDC Check

B · 15 MIN · THE SWAP

A pot of 100 ETH and 300,000 USDC, and your order

Choose the direction and the amount. Compare the average price obtained with 3,000: the gap is the price impact, plus the fee.

Direction

buy ETH with USDC

Amount paid in (USDC or ETH)

30000

9.0661	3,309.03	10.30 %	3,629.01
ETH RECEIVED	AVERAGE PRICE OBTAINED, USDC PER ETH	GAP WITH 3,000 (IMPACT + FEE)	POT PRICE AFTER THE SWAP

Sell **10 ETH** into the pot (100 ETH, 300,000 USDC, 0.3% fee). How many USDC do you receive, to the nearest unit?

… USDC Check

Buy ETH with **30,000 USDC**. How many ETH do you receive, to two decimals?

… ETH Check

C · 15 MIN · THE DEFI CAPSTONE

A complete case, and the module certificate

The DeFi capstone ↗ chains deposit, borrowing, swap and liquidation on a single wallet, and issues a certificate at the end. The course "How DeFi works" ↗ goes over the theory if needed.

The DeFi capstone, through to the certificate.

Finished, certificate obtained | Started | Not yet

1 of 1

✍️ **Would you place your customers' pension savings in a DeFi lending pot paying 4%? Three reasons, each taken from a mechanism seen this morning.**

e.g. No: 1) the yield comes from borrowers who can be liquidated in a cascade… 2) … 3) …

SECTION 4.5 · BLOCK 2

✅ QUIZ · 10 MIN

End-of-morning quiz

INTERACTIVE QUIZ · COUNTS TOWARDS PARTICIPATION

Eight questions, one attempt each, all explained

Launched in class; this page unlocks by itself. On your own, without an AI assistant. A missed quiz scores zero, unless arranged in advance.

🔒 **The quiz starts together.** It is launched in class; this page unlocks by itself within seconds, without reloading.

0 / 8 answered

1. A transaction is signed with Awa's private key. This proves…

- that Awa is its author, whatever happens
- that the holder of the private key authorised it, and that it has not been altered
- that the transaction is finalised
- that the recipient is identified

2. Why can a transaction in an old block not be altered quietly?

- Because the blocks are encrypted
- Because FINMA keeps a copy
- Because each block contains the previous one's fingerprint: everything after it would change, and the network would reject it
- Because transactions are anonymous

3. A simple transfer uses 21,000 gas, at a price of 2 gwei. The fee is…

- 0.000042 ETH
- 0.042 ETH
- 42 ETH
- 21,000 ETH

4. A DeFi loan has a health factor of 1.65. If the collateral's price falls by 40%, the factor becomes…

- 1.25: healthy
- 0.99: liquidatable
- 1.65: it does not depend on the price
- 0.40

5. On an x × y = k AMM, buying a lot of ETH at once…

- costs the quoted price, plus 0.3%
- costs a higher average price, because each ETH removed makes the next one dearer
- lowers the ETH price in the pot
- is refused beyond 10% of the pot

6. In March 2023, USDC fell below 0.90 dollar for a weekend. Why?

- A bug in its smart contract
- Part of its reserves was stuck at the failed Silicon Valley Bank
- Its algorithm gave way, like TerraUSD's
- The United States had banned it

7. According to Wüst and Gervais's tree, if an always-available trusted third party can keep the ledger…

- a public blockchain is needed
- a private blockchain is needed
- a blockchain is not needed: the third party does it better, faster and cheaper
- a stablecoin is needed

8. In a "sandwich" attack, the bot makes money because…

- it knows your private key
- your order is visible in the public queue before it is executed, and it places a buy in front and a sell behind
- it hacks the AMM's contract
- it is always the validator of the next block

SECTION 4.6 · BLOCK 3

🎤 TALK · 20 MIN

Tokenisation and central bank digital currencies

The instructor's talk, or read on your own.

TOKENISING

A token is a ledger entry that represents a right

Tokenising a building, a bond or a fund unit means recording the ownership right in a shared ledger rather than in a custodian's. What you gain: transfers settled in minutes instead of two days, smaller units, rules (who may buy, when the coupon is paid) written into the token. What you do not gain: **the blockchain does not know that the building exists**. The link between the token and the asset remains a contract, a company, a custodian and an applicable law — exactly as for a share.

SWITZERLAND

A law, an exchange, a wholesale currency

- **The DLT Act** (the federal act adapting federal law to developments in distributed ledger technology, in force in 2021) creates **ledger-based securities** (art. 973d CO): a security that exists only in an electronic register, transferable on that register with the same legal certainty as a paper security. It also creates a licence for a **DLT trading facility** and protects clients' crypto-assets if the custodian goes bankrupt.
- **SDX** (SIX Digital Exchange), the SIX group's digital exchange, authorised by FINMA in 2021, issues and settles digital bonds.
- **Helvetia**: the SNB makes a **wholesale central bank digital currency** available to banks on SDX's platform, to settle tokenised bonds in central bank money. In pilot operation since December 2023, extended by the SNB until at least June 2028. A digital currency **for the public** (retail) is not planned by the SNB.

WAEMU

Studies, and no decision

- **A BCEAO digital currency?** On 22 July 2026, presenting its annual report, the BCEAO said that a study phase is under way and that **no deployment decision has been taken**. The “eCFA” launched in Senegal in 2016 was not central bank money: it was electronic money issued by a commercial bank, on a start-up’s technology.
- **Crypto-assets.** The BCEAO points out that they are not money and are not regulated in WAMU; a regulatory framework is being prepared. Neighbouring Nigeria launched a retail digital currency, the eNaira, in 2021: its adoption has remained very low.

CENTRAL BANK DIGITAL CURRENCY: RETAIL, WHOLESALE, OR NOT AT ALL?

Helvetia: digital francs from the SNB to settle tokenised bonds on SDX

Retail CBDC | Wholesale CBDC | Not a CBDC

The Central Bank of Nigeria’s eNaira (2021)

Retail CBDC | Wholesale CBDC | Not a CBDC

The “eCFA” launched in Senegal in 2016

Retail CBDC | Wholesale CBDC | Not a CBDC

USDC

Retail CBDC | Wholesale CBDC | Not a CBDC

The digital euro being prepared at the ECB

Retail CBDC | Wholesale CBDC | Not a CBDC

A TWINT payment

Retail CBDC | Wholesale CBDC | Not a CBDC

0 of 6 sorted

SECTION 4.7 · BLOCK 3

🛠 WORKSHOP · 55 MIN

Split a building · do you need a blockchain?

A · 20 MIN · THE FRACTIONALISATION SIMULATOR

A building in Lausanne, in 1,000-franc tokens

A company owns the building; its equity is split into tokens. Change the value, the debt, the rent: what one token is worth, what it yields — and what it loses when the building loses 10%.

Building value, CHF

12000000

Mortgage, CHF

4000000

Number of tokens

8000

Rent net of costs, CHF per year

360000

Mortgage rate, %

2

1,000 CHF
VALUE OF ONE TOKEN

35.00 CHF
PAID OUT PER TOKEN PER YEAR (3.50%)

850 CHF
ONE TOKEN IF THE BUILDING LOSES 10% (-15.0%)

Building 12 million, mortgage 4 million at 2%, net rent CHF 360,000, 8,000 tokens. How much does one token **pay out per year**, in CHF?

… CHF Check

The building loses 10%. What is **one token** worth, in CHF?

… CHF Check

✍ **Could your customers' fund units be tokens (ledger-based securities)? What the customer would gain, and what would not change.**

e.g. Gained: the customer sees their unit in a public register… Unchanged: a Swiss custodian, the fund and the exchange rate are still needed…

B · 20 MIN · BY TEAM · “DO YOU NEED A BLOCKCHAIN?”

Six questions, applied to your app

The method is published: Wüst and Gervais, *Do you need a Blockchain? ↗* (2018). Take one precise function of your app — the register of customers’ units, the monthly statement, the transfer of payments from Africa — and answer for it. Start again for a second function.

The function of your app you are testing

e.g. the register of custo

Do you need to store data (a state: balances, a register)?

Yes | No

✍ **For each function tested: the tree’s result, and your decision argued in two sentences. This is the dossier’s “argued technology choice”.**

The tree inserts its result; add your decision.

DEBRIEF · 15 MIN

Each team reads its two results. Where did the tree say “no blockchain”? What tipped the answer — an available trusted third party, known actors? And if the answer is “no”, what from today is still useful for your app? Section 4.9 gives part of the answer.

SECTION 4.8 · BLOCK 4

🎤 TALK · 15 MIN

MEV, three hacks, MiCA

The instructor's talk, or read on your own.

THE "SANDWICH" ATTACK

Your order is public before it is executed

A transaction waits in the **public queue** (the *mempool*) before it enters a block, and whoever builds the block chooses the order. A bot sees your purchase of 30,000 USDC of ETH on the pot from section 4.4. It **buys just before you** (the price rises), your purchase goes through at the higher price, and it **sells just after**. It gains exactly what you lose, within your **slippage tolerance**: with 1%, you agree to receive 1% less than expected, and that is what the bot takes. This gain that the order of transactions makes extractable is called **MEV** (*maximal extractable value*).

In this example the bot puts 1,586 USDC in front of you, you receive 8.9754 ETH instead of 9.0661, and it sells back with **a 318 USDC gain**, pot fees included. The simulator "The dark forest" ↗ shows it block by block.

With no attack, 30,000 USDC buy 9.0661 ETH. With a 1% slippage tolerance, what is the **minimum ETH** you agree to receive, to two decimals?

… ETH Check

THREE HACKS

The code, the keys, the screen

- **The DAO, June 2016 — the code.** An investment fund run by a smart contract. A flaw allowed a withdrawal to be requested again before the contract had updated the balance (*re-entrancy*): about 3.6 million ETH diverted. Ethereum rewrote history through a split (*hard fork*); those who refused it carried on under the name Ethereum Classic. Lesson: "code is law" applies to bugs too.
- **Ronin, March 2022 — the keys.** The bridge of the game Axie Infinity required 5 signatures out of 9 validators. The attackers obtained 5 keys and withdrew about 625

million dollars; the FBI attributed the attack to North Korea. Lesson: a secure chain does not protect badly kept keys.

- **Bybit, February 2025 — the screen.** The platform signed its transfers from a multi-signature “cold” wallet. The signing interface had been hacked: the signers approved what they saw on screen, not what they were signing. About 1.5 billion dollars of ETH, the largest theft in crypto-asset history, also attributed to North Korea. Lesson: check what you sign, on a second device.

▸ THE CODE · THE KEYS · THE SCREEN: WHERE DOES EACH RISK COME FROM?

The DAO (2016): a withdrawal requested again before the balance was updated

The code | The keys | The screen, the person

Ronin (2022): 5 validator keys out of 9 in the attackers’ hands

The code | The keys | The screen, the person

Bybit (2025): signers approve what the interface shows them, not what they sign

The code | The keys | The screen, the person

A student gives their secret phrase to a fake “MetaMask support”

The code | The keys | The screen, the person

A platform keeps all its customers’ keys on one server connected to the Internet

The code | The keys | The screen, the person

0 of 5 sorted

THE FRAMEWORK

MiCA in the EU, FINMA in Switzerland, the BCEAO

- **European Union · MiCA** (Regulation 2023/1114). Since 30 June 2024 for stablecoins, since 30 December 2024 for the rest: an issuer of a stablecoin backed by a single

currency (an *e-money token*) must be a credit or e-money institution, with segregated reserves and a right to redemption at par; a provider (platform, custodian, adviser) needs an authorisation, valid across the Union. The national transitional periods ended on 1 July 2026 at the latest.

- **Switzerland.** No MiCA: FINMA applies existing law according to the token's function (payment, investment, utility), the DLT Act, and the AMLA to every provider that transfers crypto-assets on behalf of others.
- **WAEMU.** No framework specific to crypto-assets so far (see section 4.6).

SECTION 4.9 · BLOCK 4

PROJECT AND GAME · 75 MIN

Prove the money is there · DeFi Millionaire

A · 15 MIN · THE PROOF OF INCLUSION

Each customer checks that their units are counted, without seeing anyone else's

After FTX's bankruptcy (November 2022), platforms began publishing a "proof of reserves". Its "liabilities" half is a **Merkle tree**: the fingerprint of each account (identifier, balance, and a secret *salt* that prevents guessing the others' balances) is combined two by two up to a single fingerprint, the **root**, published on the same day for everyone. Each customer receives their path — three fingerprints here — and recomputes the root themselves. If their balance had been left out or reduced, the root would not match. Here are eight customers of the app, in fund units.

› THE MERKLE TREE OF THE EIGHT CUSTOMERS · SHA-256 COMPUTED IN YOUR BROWSER

You are customer

c03

Root published by the company

3ab7213e6ca36165…

Check my inclusion

What if the company had reduced my balance?

The root proves that your balance is *counted*. It does not prove that the units exist: that is the other half — an attestation from the custodian and the auditor that the assets held cover the sum of the balances. A blockchain can serve to **publish the root** with a timestamp and unalterably; it replaces neither the custodian nor the auditor.

B · 30 MIN · BY TEAM · THE PROJECT

Prove that your customers' money is there

1. What your customer can check themselves, without taking your word for it: what, by what means (custodian's statement, published root, token register…), how often — and who attests the other half (the assets).

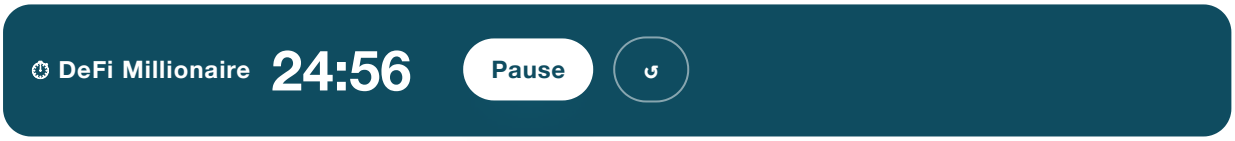
e.g. Every month the Merkle root of the balances is published… ; the Swiss custodian attests… ; the auditor…

2. The lesson of the three hacks for your app: who can move customers' money, with how many signatures, on which devices — and what happens if an employee or a phone is compromised.

e.g. No employee can move the units: only the custodian, on instruction… ; two signatures out of three, checked on a second device…

C · 25 MIN · DEFI MILLIONAIRE, BY TEAMS

The class game DeFi Millionaire ↗ is played on the room's screen, both teams at once. The questions go back over the day.



BEFORE DAY 5 · SATURDAY 31 OCTOBER

- Day 5 is a **working day**: by 16:45 each team delivers a web page with the full plan of its app — customer, fund and fees, path of the money, technology, regulation, unit economics, funding — built with an AI, checked by the team, published and handed in on this site.
- Bring your fields from Days 1 to 4 and a Claude or ChatGPT account. For the project: note which regulator, in your customer's country and in Switzerland, would have to authorise your app — Day 5 makes you choose the structure.

DAY 5 · SATURDAY 31 OCTOBER 2026 · MFB 545

Workshop day · your app's full plan, on one web page

No lecture today: you build. By 16:45 each team has published **one web page** presenting the full plan of its pension-savings app — the customer, the fund and its fees, the path of the money from Dakar or Brazzaville to Zurich, the technology, regulation here and there, the company's numbers, the funding — and handed it in on this site. You will use an AI (Claude, ChatGPT…) to write and to code. You are graded on what the AI cannot do: choose, quantify, verify.

🕘 **09:00–12:15 · 13:30–16:45 · four blocks** | 💻 **one computer per person · Claude or ChatGPT open**

🏁 **16:45: the page is online and handed in on rendu.html**

TODAY'S DELIVERABLE

One web page per team (a single `index.html` file), published at an address anyone can open, with the twelve parts of section 5.6. It becomes the skeleton of the 20 November dossier and of the 14 November presentation. Section 5.6 gives the method step by step; 5.7, the grid the other team will review you with at 16:00.

The brief · section 5.1 | **Build and publish the page · 5.6** | 📤 **Hand in**

BY THE END OF DAY 5, YOUR TEAM WILL HAVE

- A precise customer, a fund that exists (with its ISIN and real fees), and the capital at 65 computed — with the share taken by fees.
- The path of the money from the customer's phone to the fund units, every intermediary and every fee named, and the list of what you buy and what you build.
- An argued answer to "should this be a blockchain?" and the proof you offer the customer that their money is there.
- The regulatory map — in Switzerland, in WAEMU or CEMAC — and the legal structure chosen to hold within it.
- Unit economics: what a customer brings in per year, what one costs to acquire, how many you need to live.

- The funding need and three milestones.
- A published web page, reviewed by the other team, handed in on this site.

BEFORE YOU START — TICK

- ☐ **Signed in** on this site (the button at the bottom right), and **in your team** (course page).
- ☐ **Claude or ChatGPT** open in a tab, with an account (free is enough).
- ☐ **Your fields from Days 1 to 4** at hand: the workbook (course page, “Workshop notebook”) gathers them — customer, flow, revenue, plan, transparency, safeguards, proof, keys.
- ☐ **A calculator**: every number on the page is recomputed by you before it goes on.

BLOCK 1 · 09H00–10H30 the brief · the customer and the product

5.1 **What you deliver, and the rules of the game with the AI**

One web page, twelve parts · five screens · the AI writes, you verify · the roles

🎤 TALK · 30 MIN

5.2 **The customer and the product · screens 1 and 3**

A precise customer · a fund with its ISIN · the projector and the fee share

🛠 WORKSHOP · 60 MIN

BLOCK 2 · 10H45–12H15 the flow of money and the technology · the page takes shape

5.3 **The flow of money and the technology · screens 2 and 4**

The chain of hands and its fees · buy or build · blockchain or not, the proof

🛠 WORKSHOP · 75 MIN

BLOCK 3 · 13H30–15H00 regulation and unit economics

5.4 **Regulation and unit economics · screen 5 and the company**

FINMA, AMF-UMOA, BEAC, CIMA · the capital outflow · what a customer brings in, the break-even

🛠 WORKSHOP · 75 MIN

BLOCK 4 · 15H15–16H45 funding · build and publish · cross-review and hand-in

5.5 **Funding and the roadmap**

How much, for what, three dated milestones

🛠 WORKSHOP · 25 MIN

5.6 **Build and publish the page · step by step**

The twelve parts · the prompt · eight steps to the public address

🛠 WORKSHOP · 35 MIN

5.7 **The other team reviews your page · then you hand in**

Ten minutes on screen · the “can I check this?” grid · the hand-in on rendu.html

🔁 DEBRIEF · 45 MIN

← Course page

reset my Day 5 progress on this device

SECTION 5.1 · BLOCK 1

🎤 THE BRIEF · 30 MIN

What you deliver, and the rules of the game with the AI

Read together at 9:00. Then each team shares out the roles and fills in the field at the bottom of the page.

THE DELIVERABLE

One web page, twelve parts, online at 16:45

Your app's full plan fits on **one web page** — an HTML file you will have an AI write from your texts and your numbers, then publish at a public address and hand in on rendu.html, piece "The plan page". The twelve parts are fixed (section 5.6): they follow the order of the graded dossier of 20 November, so today's page is its first draft, and the first draft of the 14 November presentation.

The page is not the prototype. The prototype shows the screens; the page explains *what is behind each screen*: who, how much, through where, who authorises, who holds the money, who pays what.

THE METHOD

The app is five screens. The plan is everything behind them.

To avoid writing a thesis, start from the screens. A customer opens the app and sees five things. Behind each, a part of the plan — and a day of the course.

SCREEN	WHAT THE PLAN MUST SAY BEHIND IT	SEEN IN
1 · I open an account	Who the customer is, which documents for KYC, in how many minutes, with which identity provider — and what is refused.	Day 1 — eKYC, buy or build
2 · I pay in	Wave, Orange Money, M-Pesa or transfer; what a 10 000 FCFA payment costs at each hand on its way to Switzerland; the exchange.	Day 2 — the rails, interchange, Geneva → Dakar
3 · My money	The balance, fees in francs, performance, the exact fund with its ISIN; the sentence for the day the fund has lost 20%.	Day 3 — the projector, transparency

4 · I verify	The custodian's statement, the published root, who attests the assets; how many signatures to move the money.	Day 4 — proof of reserves, "should this be a blockchain?"
5 · I withdraw	At 65 and before; what it costs; what the country's law and Swiss law allow — and who authorises you to offer this.	Today — regulation, section 5.4

And behind the five screens, the company: what it buys, what it builds, what a customer brings in, what one costs, what it must raise.

THE RULES OF THE GAME

The AI writes. You decide, you quantify, you verify.

> Claude or ChatGPT drafts the paragraphs, proposes names, codes the page. **You are graded on what the AI cannot do**: the fund exists and its ISIN checks out; the fees come out of this site's projector; the transfer cost is the one you observed on Day 2; the regulator is named with its text; the break-even is your arithmetic. **Every number on the page carries a "source: …" line**. At 16:00 the other team reviews your page with one question per part: *can I check this?* A claim without a source counts for zero.

Both teams have the same brief. The only way to stand out is to be **precise**: a customer with a first name, an income and a phone; one fund; one payment rail; one legal structure; a launch date.

› WHO DOES WHAT? SORT EACH TASK OF THE DAY

Draft the paragraph presenting the customer from your notes

The AI writes it | You do it, you verify it

Check that the fund's ISIN exists and read its TER on today's factsheet

The AI writes it | You do it, you verify it

Compute the capital at 65 and the share taken by fees

The AI writes it | You do it, you verify it

Propose five names for the app and a tagline

The AI writes it | You do it, you verify it

Name the regulator of the customer's country and the text that applies to the capital outflow

The AI writes it | You do it, you verify it

Write the HTML and CSS code of the page

The AI writes it | You do it, you verify it

Look up what Wave or Orange Money takes on a collection, and the exchange price

The AI writes it | You do it, you verify it

Summarise WAEMU regulation 06/2024 in five lines

The AI writes it | You do it, you verify it

0 of 8 sorted

ROLES · 5 MIN · PER TEAM

Three people, three workstreams, one page

Customer and product
Screens 1 and 3. The customer, the fund and its ISIN, the projector, fees in francs, withdrawal. Sections 5.2 and the start of 5.4.

Money and compliance
Screens 2, 4 and 5. The chain of hands and its fees, buy or build, the proof, the regulators and the structure. Sections 5.3 and 5.4.

Numbers and page
Unit economics, funding, then the page: the prompt, the file, publishing, the hand-in. Sections 5.4 (B), 5.5 and 5.6.

Everyone writes in their own workbook — **the same version for the whole team**: when a part is ready, the person holding the page pastes it into the file.

1. Your team: the customer's country and city, the app's working name, and who holds which workstream (three first names).

e.g. Senegal, Dakar — app "Teranga Retraite" — Awa: customer and product; Moussa: money and compliance; Fatou: numbers and page

SECTION 5.2 · BLOCK 1

🛠 WORKSHOP · 60 MIN

The customer and the product · screens 1 and 3

A · 15 MIN · THE CUSTOMER

A person, not a target group

In Senegal the informal economy is close to 90% of activity and only about 20% of its workers had pension coverage before the agreements signed in 2026 to bring them into IPRES (source: Le Soleil, May 2026 — a figure to quote with its source, not as a certainty). The self-employed and informal workers contribute to a pension only on a voluntary basis (OECD, *Financement de la protection sociale au Sénégal*). Your customer is one of these people. Give them a first name, an age, a trade, a monthly income, a phone, and say **what they do today for their old age**: the tontine, cash at home, a plot of land, the children. Then what would make them entrust 10 000 FCFA a month to an app rather than the tontine.

Documents: which ID do they actually have (ECOWAS national identity card, passport, none)? Your Day 1 KYC must work with *that* document, in under ten minutes, on *that* phone.

1. The customer: first name, age, trade, monthly income and what they can pay in, phone and connection, ID document, what they do today for their pension — and the reason that would make them try your app.

e.g. Khady, 32, trader at Sandaga market, about 250 000 FCFA a month, can put aside 10 000; Android smartphone, Orange Money; ECOWAS ID card; today a 25 000-a-month tontine and a plot in Thiès; would try if…

B · 15 MIN · THE FUND

A fund that exists, with its ISIN and real fees

The brief says “Swiss blue-chip equity funds”. Those are not just words: it is a listed product, with a twelve-character ISIN code, annual fees (the *TER*) and a factsheet. Two

examples tracking the SMI index of the twenty largest Swiss companies, both Swiss-domiciled and distributing:

FUND	ISIN	FEES (TER)	SIZE
UBS ETF (CH) SMI	CH0017142719	0.20% a year	≈ CHF 3.8 billion (justETF, July 2026)
iShares SMI ETF (CH)	CH0008899764	0.35% a year	≈ CHF 2.6 billion (justETF, July 2026)

Check for yourselves before quoting: type the ISIN on justetf.com ↗ or on the issuer's site and read the TER on today's factsheet. A fund whose ISIN cannot be found does not exist for your page. You may choose another (SPI index, dividend fund, sustainable pension fund…), on the same condition.

▸ THE FUND'S COST, IN FRANCS

A customer holds 100 000 FCFA of units in the UBS fund (TER 0.20%). How much does the fund cost them per year, in FCFA?

… FCFA Check

2. The fund chosen: name, ISIN, TER, what it holds, where you checked it — and in one sentence why this one rather than the other.

e.g. UBS ETF (CH) SMI, CH0017142719, TER 0.20%, the 20 largest Swiss companies (Nestlé, Novartis, Roche…), checked on justetf.com on 31 October 2026; chosen because…

C · 30 MIN · THE PLAN AND ITS FEES

What a monthly payment becomes at 65 — and what the fees take

Day 3's projector, with your settings (they are kept). Enter **your** customer, **your** fund, **your** fees. The page must show these five numbers to the customer, and the share taken by fees in plain words.

▸ THE PENSION PROJECTOR

Payment per month

10000

Customer's currency

FCFA

Age at first payment

30

Annual fund return, before fees (%) — an assumption

5

Fund fees, % per year

0.2

Your app's fee, % of assets per year

0.8

Transfer and exchange, % of each payment

1

Swiss franc against the customer's currency, % per year

0

4,200,000 FCFA
PAID IN FROM 30 TO 65

8,846,806 FCFA
THE CAPITAL AT 65, AFTER ALL FEES

11,129,790 FCFA
THE SAME PLAN WITH NO FEES AT ALL

20.5 %
THE SHARE OF THE CAPITAL TAKEN BY FEES

6,192,764 FCFA
IF EQUITIES LOSE 30% IN THE YEAR YOU TURN 65

Payments at the start of each month, return compounded monthly, annual fees deducted from the return. The return is an assumption, not a promise: Swiss blue chips also lost more than a third in 2008 and about a quarter in a few weeks in 2020. The Swiss franc gained against the euro, hence against the CFA franc, over twenty years (about 1.60 francs per euro in 2007, 0.93 in 2025), but nothing guarantees what comes next. These are the numbers the page shows the customer.

› READ THE PROJECTOR

With the default settings — 10 000 FCFA a month from age 30, 5% a year, fees 0.2% + 0.8% a year and 1% per payment — what share of the capital at 65 do the fees take, in %?

… % Check

3. The plan in numbers: payment and currency, starting age, the capital at 65 after fees and without fees, the fee share in %, each fee in francs on a 10 000 payment; the payout at 65 (lump sum, annuity, by Wave?); early withdrawal: allowed or not, delay, cost.

e.g. 10 000 FCFA a month from age 30 → 8 846 806 FCFA at 65 after fees (11 129 790 without fees, fees take 20.5%); on 10 000: 100 for transfer and exchange, …; at 65: …; early withdrawal: …

SECTION 5.3 · BLOCK 2

🛠 WORKSHOP · 75 MIN

The flow of money and the technology · screens 2 and 4

A · 25 MIN · THE CHAIN OF HANDS

From 10 000 FCFA on a phone to a fund unit in Zurich

Every hand that touches the money has a name, a licence and a price. Here is the most common path; your page repeats it with **your** providers and **your** fees, observed and not guessed.

1 · The customer's phone. Khady opens the app and confirms 10 000 FCFA. The app does not touch the money: it sends a collection order to the mobile-money operator (Wave, Orange Money, MTN MoMo, M-Pesa), licensed by the central bank as an *e-money issuer*.

2 · The mobile operator collects. It debits Khady's mobile account and credits your company's **collection account**, held at a local bank. It takes its merchant commission — to be read on its public price list, usually between 0 and 2% depending on operator and volume.

3 · The local bank pools. At each month end, all customers' payments are pooled into **one international transfer**: this is what makes small amounts possible — one SWIFT transfer per customer would cost more than the payment.

4 · The capital outflow is authorised — or not. The money leaves WAEMU or CEMAC. The exchange regulation (section 5.4) requires an authorisation: either per customer (impractical), or for the structure carrying the funds (a licensed collective scheme, an insurer). This is where your choice of structure is decided.

5 · The exchange. CFA franc → euro at the fixed rate (655.957), then euro → Swiss franc at the day's rate, at a bank or an exchange provider (Wise Business, a Swiss bank). Cost: the margin on the rate, 0.3 to 1%. The plan's exchange risk is CHF/EUR, and nothing else.

6 · The Swiss custodian buys the units. A custodian bank in Switzerland receives the francs on an account **segregated** from your company's accounts and buys the fund units — in each customer's name, or on an omnibus account with a register per customer (your choice, and it changes what the customer can verify).

7 · The statement. The custodian publishes the attestation of units held; your app shows Khady her balance, her fees in FCFA and her performance, and — if you decided so in 5.3 C

— her Merkle path. The money took between one day and five weeks to become a fund unit.

✓ All 7 steps are shown

▸ WHAT ARRIVES IN SWITZERLAND

Khady pays in 10 000 FCFA. The mobile operator takes 1%, the local bank a fixed 500 FCFA per batch of 100 payments, the exchange 0.5% of what remains. How much reaches the Swiss custodian's account, in FCFA?

… FCFA Check

The CFA franc (XOF and XAF alike) is pegged to the euro: 655.957 CFA francs per euro, without variation. The Swiss franc floats against the euro — around 0.93 francs per euro in 2025, so about 705 FCFA per franc. The only exchange-rate uncertainty in your plan is therefore CHF/EUR; say it that way on the page.

1. Your chain of hands: each intermediary named (mobile operator, local bank, exchange and transfer provider, Swiss custodian, fund issuer), what each takes on 10 000 FCFA, what reaches the fund, and how long it takes.

e.g. Orange Money (1% merchant collection) → collection account at … in Dakar → pooled transfer at each month end → FCFA/CHF exchange at … (0.4%) → custody account at … in Zurich → units bought on the 5th; … FCFA out of 10 000 arrive; delay …

B · 20 MIN · BUY OR BUILD

The unbundled bank, applied to you

Day 1 showed a bank in separate pieces. Your company builds only a few of them; the rest is rented from providers who already have the licence and the customers. Sort, then write on the page the list of what you buy (and from whom) and what is yours.

▸ BOUGHT OR BUILT?

Identity verification (reading the document, face matching, sanctions-list screening)

Bought or rented from a provider | Built by your company

Custody of the fund units

Bought or rented from a provider | Built by your company

The fund itself

Bought or rented from a provider | Built by your company

Collection by mobile money

Bought or rented from a provider | Built by your company

The app and its five screens

Bought or rented from a provider | Built by your company

The monthly statement and the inclusion proof

Bought or rented from a provider | Built by your company

The customer relationship — the call when a payment failed or when the fund fell

Bought or rented from a provider | Built by your company

Anti-money-laundering transaction monitoring

Bought or rented from a provider | Built by your company

0 of 8 sorted

C · 30 MIN · THE TECHNOLOGY AND THE PROOF

Should this be a blockchain? And how the customer checks they are counted

Day 4 gave the method (section 4.7) and the inclusion proof (section 4.9). Decide for your app — three choices, recorded, that go on the page as they are.

▸ THREE DECISIONS

A blockchain in your app?

No — an ordinary database | Yes, to publish the root of the balances | Yes, the units are tokens

Who attests that the units exist?

The custodian, every month | The custodian + an auditor, every quarter

Each customer, through the inclusion proof + the custodian's attestation

Who can move the customers' money?

Nobody at your company — only the custodian, on written instruction

Two signatures out of three at your company, on separate devices | One authorised person

0 of 3

2. Your technology choice in one paragraph: what is an ordinary database, what (if anything) goes through a blockchain and why, what the customer checks themselves each month, who attests the assets — and who can move the money, with how many signatures.

e.g. No blockchain for the accounts: a database at …; each month the Merkle root of the balances is published on …; the custodian … publishes the attestation of units; the auditor …; no single employee can…

SECTION 5.4 · BLOCK 3

🛠 WORKSHOP · 75 MIN

Regulation and unit economics · screen 5 and the company

A · 40 MIN · THE REGULATORY MAP

Who must authorise you — here, and there

Three sets of rules meet in your app: the customer's country (their money leaves the country), Switzerland (the fund and its custody are there), and the rules that apply to anyone handling other people's money (money laundering). Read the four cards, check each text at its source before quoting it, then choose the structure.

SWITZERLAND · THE FUND, CUSTODY, ADVICE

FINMA · FinSA · FinIA · CISA · AMLA

Offering a collective investment scheme to retail clients makes you a **financial service provider** under FinSA: advisers entered in a register, affiliation to an ombudsman, duties of information and appropriateness checks. **Managing** client assets (deciding the purchases) requires a **portfolio manager** licence (FinIA), supervised by a supervisory organisation approved by FINMA (OSIF, OSFIN…). A fund offered in Switzerland to retail clients must be **approved** (CISA). Collecting and forwarding other people's money makes you a financial intermediary under the **AMLA**: affiliation to a self-regulatory organisation, customer identification, transaction monitoring. **Check on finma.ch before quoting.**

WAEMU · SENEGAL, CÔTE D'IVOIRE, MALI, NIGER, BURKINA, TOGO, BENIN, GUINEA-BISSAU

AMF-UMOA · BCEAO · the exchange regulation

The **AMF-UMOA** (the WAMU financial markets authority — the former CREPMF, renamed on 1 October 2022) licenses management companies, UCITS and any public offering of investments in the Union. The **BCEAO** licenses e-money issuers and payment providers. And above all: **Regulation No. 06/2024/CM/UEMOA** of 20 December 2024 (replacing 09/2010) makes **any investment abroad by a resident subject to the prior authorisation of the Minister of Finance**. A Dakar resident buying units of a Swiss fund every month is making an investment abroad. **This is the question that decides your structure.**

CEMAC · CONGO-BRAZZAVILLE, CAMEROON, GABON, CHAD, CAR, EQUATORIAL GUINEA

COSUMAF · BEAC · regulation 02/18

The **COSUMAF** supervises the regional financial market and licenses collective

THE INSURANCE ROUTE

CIMA · a retirement-savings contract

In the fourteen **CIMA** countries (the Inter-African Conference of Insurance Markets — WAEMU and CEMAC together, plus others), a licensed life insurer may sell a

schemes and their managers. The **BEAC** applies **Regulation No. 02/18/CEMAC/UMAC/CM** of 21 December 2018 (in force 1 March 2019) and its instruction of 5 October 2020 on direct and portfolio investments with abroad: residents' capital outflows are subject to authorisation and supporting documents, and the rules have been enforced strictly since 2019. **DRC** (Congolese franc, outside CEMAC): the Central Bank of Congo has its own exchange regulation. **Check on beac.int and cosumaf.org.**

unit-linked **retirement-savings contract**. The insurer carries the licence, the custody and the relationship with the regulator; your app is the distribution channel and the interface. The insurers' investment rules (share allowed outside the zone) then apply to the Swiss fund. **CIMA Code, Book IV, to be checked.**

▸ YOUR STRUCTURE · ONE DECISION, RECORDED

How does the customer's money legally reach a Swiss fund?

- Each customer buys units of the Swiss fund themselves
- A licensed local vehicle (collective scheme, management company) that itself invests in Switzerland
- A retirement-savings contract carried by a licensed life insurer (CIMA)
- The customer is a Swiss resident (the diaspora) saving for a relative back home

0 of 1

1. Your regulatory map: the structure chosen and why; in Switzerland, the licences or affiliations needed (named); in the customer's country, the regulator and the text that apply; how you handle the capital outflow; KYC/AML; what you do not know yet and who will tell you.

e.g. Structure: a retirement-savings contract carried by insurer … (CIMA-licensed in Senegal), invested in fund … via …; in Switzerland the company is affiliated to …; regulation 06/2024 is respected because …; KYC by …; to confirm with a lawyer: …

B · 35 MIN · UNIT ECONOMICS

What a customer brings in, what one costs, how many you need

A customer at 10 000 FCFA a month brings in little to whoever takes 1% of the payment and 0.8% of assets: this is a business of **volume** and **patience** — assets, and so fees, grow with the years. Do the two calculations below, then your own with your fees, your acquisition cost (Day 3: CAC, LTV, churn) and your fixed costs.

TWO REFERENCE CALCULATIONS

A customer pays in 10 000 FCFA a month; you take 1% of each payment and 0.8% a year on their holding, on average 400 000 FCFA this year. What do they bring you over the year, in FCFA?

… FCFA Check

Your fixed costs are 210 000 000 FCFA a year (about CHF 300 000). At 4 400 FCFA per customer per year, how many customers do you need to cover them?

… customers Check

Assumptions of the two calculations: 10 000 FCFA a month, 1% per payment, 0.8% a year on an average holding of 400 000 FCFA (a customer saving for three years), fixed costs of 210 000 000 FCFA a year — about CHF 300 000 at 700 FCFA per franc: three salaries, compliance, hosting, support. Change all of it for your case; keep the method.

2. Your numbers: revenue per customer per year (in year 1 and year 5), cost to acquire a customer and through which channel, assumed churn, annual fixed costs, number of customers at break-even — and the year you reach it.

e.g. ARPU 2 400 FCFA in year 1, 6 000 in year 5; CAC 1 500 FCFA through Orange Money agents; churn 15% a year; fixed costs 180 million; break-even 40 000 customers, reached in year 4 if…

SECTION 5.5 · BLOCK 4

🛠 WORKSHOP · 25 MIN

Funding and the roadmap

25 MIN · PER TEAM

How much, for what, to which milestone

An investor will read your page with three questions. **How much are you raising** and to last how many months? **What do you buy with it** — the licences and affiliations of section 5.4 cost money and time before the first customer; the app; acquisition? **Which milestone** will prove it works — a pilot of 500 savers paying three months in a row, the licence obtained, 50 000 customers? At 4 400 FCFA per customer per year, 50 000 customers make 220 million FCFA of revenue, about CHF 314 000: tell the investor how you get there, and in how many years.

The detail (cap table, dilution, valuation) comes on the last day. Today: three numbers and three dated milestones.

1. Funding: the amount to raise and the period it covers, the use of funds in three or four lines, three dated milestones, and what you ask the investor for besides money.

e.g. Raise CHF 450 000 for 24 months: 150 000 licences, counsel and compliance; 150 000 app and security; 100 000 acquisition and agents; 50 000 reserve. Milestones: licence … (June 2027), pilot of 500 savers (Dec. 2027), 10 000 customers (Dec. 2028)…

SECTION 5.6 · BLOCK 4

🛠 BUILD · 35 MIN

Build and publish the page · step by step

*Start this section **in the morning** for the skeleton, and come back to it as each part is finished. At 16:00 the page is online, even if imperfect.*

THE TWELVE PARTS

What the page contains, in this order

The order is that of the 20 November dossier. Each part is five to fifteen lines, at least one number, and a *source* line. The page carries a title, the app's name, the country, the three first names, and the date.

1. **The problem in one sentence**
   Who has no pension, how many they are, what they do today — with the source (5.2 A, Day 1).

2. **The customer**
   The portrait from 5.2 A: first name, age, trade, income, phone, ID document, what would make them try.

3. **The product**
   The plan: payment, age, fund with ISIN and TER, the projector's five numbers, the fee share, the payout at 65, early withdrawal (5.2 B and C).

4. **The five screens**
   One line per screen: what the customer sees, and the sentence shown the day the fund has lost 20%.

5. **The path of the money**
   The chain of hands with each fee, what reaches the fund out of 10 000 FCFA, the delay, the exchange (5.3 A, Day 2).

6. **Buy or build**
   The list from 5.3 B, with the names of the providers in mind.

7. **Technology and proof**
   Database or blockchain, and why; what the customer checks; who attests; the signatures (5.3 C, Day 4).

8. **Trust**
   Who holds the money and the units, the monthly statement, the safeguards against misappropriation, mis-selling, exchange risk, lapsed payments (Day 3, 3.9).

9 **Regulation**
The structure chosen; the licences in Switzerland; the regulator and the text in the customer's country; the capital outflow; KYC/AML; what remains to be confirmed (5.4 A).

10 **Unit economics**
Revenue per customer, acquisition cost, churn, fixed costs, break-even and its year (5.4 B).

11 **Funding and milestones**
Amount, period, use of funds, three dated milestones (5.5).

12 **Risks and what we do not know**
Five honest lines: what can make the project fail, and the open questions — the part a jury reads first.

THE PROMPT

What you give the AI so it writes the page

Copy the block below into Claude or ChatGPT, then paste your twelve parts **after it** (the text of your fields, cleaned up). Never ask the AI to invent a missing number: leave "[to complete]" and come back to fill it in.

```
You are a web developer. Write ONE complete web page, in a single HTML
file, in English, with no external library and no image to download,
presenting the plan of a pension-savings application for [COUNTRY].

Constraints:
- Everything is in the file: the CSS in a <style> tag, no external font,
no external script.
- At the top: the app's name, the country and city, the team's three first
names, "MFB 545 · Swiss UMEF · 31 October 2026".
- Twelve sections in the order below, with a menu at the top that jumps to
each. Each section has a title, my text (do not rewrite it, only fix
spelling), and a small "Source: …" line at the bottom when I give one.
- The numbers I give are reproduced exactly; if a number is missing, write
"[to complete]" in red, invent nothing.
- A table for section 3 (the plan's five numbers) and for section 5 (the
chain of hands: intermediary, what it takes, what remains).
- Readable on a phone: one column, 16 px text, 16 px margins.
- Sober: two colours, a system font, no animation.
- At the end of the file, nothing but the page. Give me the complete file,
ready to save as index.html.
```

Copy the prompt

STEP BY STEP

From the prompt to the page online, in eight steps

1 · Gather your twelve parts in one document (Google Docs, Word, a shared notepad). One part = a title + your text + “Source: …”. Copy your fields from this site (the workbook, course page). Mark what is missing “[to complete]”: that is your afternoon to-do list.

2 · Give the prompt to the AI, then your twelve parts after it, in the same message. Ask for *the complete file*. If the answer is cut off, write “continue the file from the last line”; if it rewrote your texts, “use my texts word for word”.

3 · Save the file as `index.html`. In Claude, the artifact’s download button; in ChatGPT, “Copy code”, then paste into a plain-text editor — Notepad (Windows) or TextEdit in “Format ▸ Make Plain Text” mode (Mac) — and save as `index.html`, UTF-8 encoding. Never Word: it breaks the code.

4 · Open the file in your browser (double-click). Proofread *the text*, not the code: every number, every regulator’s name, every ISIN, every “Source” line. A number the AI rounded or invented is fixed in your document (step 1), then you ask for the file again — or you fix it directly in the text editor.

5 · Put the file alone in a folder named after your app (no spaces or accents: `teranga-retraite`). The folder contains `index.html` and nothing else.

6 · Publish. Method A, the simplest: app.netlify.com/drop ↗ — drag the *folder* into the zone; within seconds the address appears (…`netlify.app`). Then create a free account (“claim”) so the address stays. To update: same page of your site, drag the folder again. Method B: a GitHub account → “New repository” → “uploading an existing file” → `index.html` → Settings ▸ Pages ▸ Source: *Deploy from a branch*, *main*, */ (root)* → the address is `yourname.github.io/reponame` after a minute or two. Method C: in Claude, “Publish” the artifact and copy the link.

7 · Test the address on another device — a phone of a member of the other team, not signed in to your accounts. The page opens, reads in one column, all the numbers are there. If it asks for an account or shows an error, go back to step 6.

8 · Save the address below, then hand it in with the `index.html` file on rendu.html, piece “The plan page” — one hand-in per team, the three first names in “Your team”. The dated

receipt is your proof of hand-in.

✓ All 8 steps are shown

YOUR PAGE'S ADDRESS

Paste the published address — the page tests it

https://…

Save

*A valid address starts with `https://` and opens on a phone **without signing in to any account**. Test it on a phone of a member of the other team before saving it. Then hand it in, with the file, on rendu.html, piece "The plan page".*

SECTION 5.7 · BLOCK 4

CROSS-REVIEW AND HAND-IN · 45 MIN

The other team reviews your page · then you hand in

16:00 · 10 MIN PER TEAM · THE PRESENTATION

The page on the room screen, and one question per part

Each team opens its page on the room screen and walks through it in ten minutes, from the problem to the funding. The other team follows with the grid below and answers, for each line, the day's only question: *can I check this?* Say your answers out loud, with a suggestion for each "no".

The page presentation **9:56** Pause ↺

THE GRID · THE TEAM YOU ARE REVIEWING

The customer is a person: first name, age, trade, income, phone, ID document, what they do today for their pension.

Yes | Partly | No

The fund exists: the ISIN is found on justetf.com or at the issuer, the TER is the factsheet's.

Checked | Not checked | No

Fees are in francs (FCFA, CDF…) and the share of the capital they take at 65 is written in plain words.

Yes | In %, not in francs | No

The chain of hands names each intermediary and each fee, and says what reaches the fund out of 10 000.

Yes | Gaps | No

The legal structure is chosen, the customer-country regulator and the capital-outflow text are named, and what remains to confirm is stated.

Yes | Vague | No

Revenue per customer, acquisition cost and break-even are computed — and the calculation can be redone in your head.

Yes | Numbers without a calculation | No

Every number on the page carries a source — and part 12 says what the team does not know.

Yes Half No

0 of 7

16:25 · 15 MIN · FIX, THEN HAND IN

What you change, and the hand-in

Fifteen minutes to fix what the grid showed — a number without a source, an unchecked fund, a vague structure. Republish (same address, same method: step 6). Then **hand in on rendu.html**, piece “The plan page”: the address *and* the `index.html` file, one hand-in per team, the three first names in the “Your team” field. The dated receipt is what counts.

1. The critique received from the other team (the “no”s and the “to check”s), what you fixed before handing in, and what remains to do for the 20 November dossier.

e.g. Two “no”s: the fee share was not in plain words, the Congo regulator was not named; fixed: …; left for the dossier: …

📤 Hand in the plan page

AFTER DAY 5

- The page is the first draft of the dossier (fifteen pages at most, Friday 20 November, 23:59) and the plan of the presentation before the jury (Saturday 14 November, 09:00). Keep the address: you will update it.
- Every “to confirm” in part 12 is a question to ask before 14 November — a regulator, an insurer, a custodian, a bank. A written answer is worth more than ten pages.